Confirming What Matters: Certified Third Normal Form Synthesis from Uncertain Functional Dependencies

Zhuoxing Zhang
University of Auckland
Auckland, New Zealand
zhuoxing.zhang@auckland.ac.nz

Sebastian Link
University of Auckland
Auckland, New Zealand
s.link@auckland.ac.nz

ABSTRACT

State-of-the-art schema design synthesizes a lossless, dependency-preserving Third Normal Form (3NF) decomposition that is optimal for given functional dependencies (FDs). Yet the FDs are estimated by mining data or asking large language models (LLMs), most mined FDs hold only by accident, and the models err. Since a single false FD can make the design lossy for the true FDs, a designer has to confirm the estimate, but cannot be asked about thousands of candidates. We propose to certify the design by a proof whose premises are the designer's answers and the data. The designer confirms a cover of the FDs that the design relies on, and refutes the maximal sets of attributes from which this cover does not derive a given attribute. We prove that the price of this certificate, one answer per FD of a smallest cover and one per such set that the data do not refute, is the length of a shortest proof. Our algorithm Cert3NF lets an advisor, built from data statistics and optionally an LLM, order the questions, so that a wrong advisor costs answers but never correctness. A pre-registered evaluation on twelve unseen relations shows that no fully automatic method we tested reliably obtains a good design, while a simulated designer certifies every design, without a budget, at about 2× to 2.5× the price in the median. Every certificate is correct where the data satisfy the true FDs, and the design is lossless for them even where a budget runs out. On synthetic schemata with up to a million candidates, the answers follow the price rather than the number of candidates. Answering every premise twice cuts the wrong certificates of a designer who errs at random from more than one half to about 6%, and in a study with nine people the experts certified all their designs. In summary, a designer's answers follow the length of the proof, however many FDs hold by accident.

PVLDB Reference Format:
Zhuoxing Zhang and Sebastian Link. Confirming What Matters: Certified Third Normal Form Synthesis from Uncertain Functional Dependencies. PVLDB, 20(1): XXX-XXX, 2027.
doi:XX.XX/XXX.XX

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/zhuoxingzhang/cert3nf.

This work is licensed under the Creative Commons BY-NC-ND 4.0 International License. Visit https://creativecommons.org/licenses/by-nc-nd/4.0/ to view a copy of this license. For any use beyond those covered by this license, obtain permission by emailing info@vldb.org. Copyright is held by the owner/author(s). Publication rights licensed to the VLDB Endowment.
Proceedings of the VLDB Endowment, Vol. 20, No. 1 ISSN 2150-8097.
doi:XX.XX/XXX.XX

1 INTRODUCTION

The logical design of relational database schemata is a classical topic of database research [12, 34]. Third Normal Form (3NF) synthesis turns a set of functional dependencies (FDs) into a lossless, dependency-preserving decomposition into 3NF [7, 10]. Recent algorithms also optimize the result, for example by minimizing the numbers of minimal keys and of non-key FDs that the schemata must maintain [53–55]. All of these guarantees hold relative to the input FDs.

Yet all approaches so far assume that the input is exactly the set Σ of business rules of the domain. In practice, Σ is estimated. FD discovery returns every minimal FD that holds on an instance [23, 39], and most of these hold by accident [48]. On the eight tables of TPC-H [44], for example, discovery returns 3,945 minimal FDs, of which only 76 follow from the specification. Large language models (LLMs) that judge FDs from the names of the attributes err frequently on real schemata. Hence, an estimate $\hat{\Sigma}$ differs from Σ, and synthesis computes the optimal design for the wrong input. A designer who knows the domain can tell genuine rules from accidental FDs, but cannot be asked about thousands of candidates. This is illustrated as follows.

Example 1.1 (Running example). Consider the schema Orders with the attributes O(rder), customer N(umber), C(ity) and D(ate). Its business rules are $\Sigma = \{O \to N, O \to D, N \to C\}$: an order has one customer and one date, and a customer lives in one city. Orders violates 3NF under these rules, since $N \to C$ holds while N is not a key, and 3NF synthesis returns the decomposition $\{ON, OD, NC\}$. On the following sample, FD discovery also reports $N \to D$, since every customer ordered on one date only.

Order	Cust. No.	City	Date
o_1	101	Auckland	3 March
o_2	101	Auckland	3 March
o_3	102	Wellington	3 March
o_4	103	Auckland	5 March
o_5	104	Wellington	3 March

On the estimate $\hat{\Sigma} = \Sigma \cup \{N \to D\}$, synthesis returns $\hat{\mathcal{D}} = \{ON, NC, ND\}$, which declares N as the key of ND. This design rejects legal data, such as a second order of customer 101 on another date. Synthesis adds no key schema, since ON contains the key O, and the design is lossless, dependency-preserving and in 3NF for $\hat{\Sigma}$, as synthesis guarantees for any input. Yet it is lossy for Σ: joining ON with ND assigns to every order of a customer the dates of all orders of this customer. In contrast, if the estimate misses $O \to D$, synthesis still returns $\{ON, NC, OD\}$, now with the key schema OD, which does not declare O as a key. Declaring it repairs the design without moving any data. □

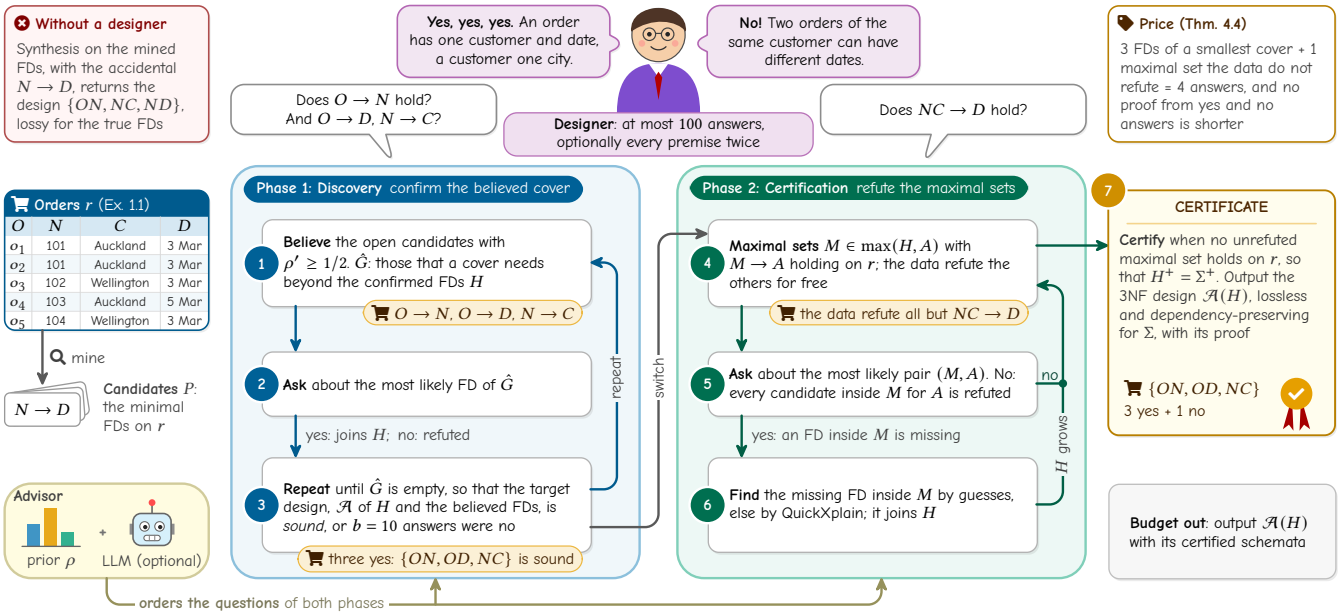


Figure 1: The Seven Steps of Cert3NF: Discovery Guided by the Advisor, Certification by the Maximal Sets

Errors thus differ in their harm, and so does what a designer must confirm. Firstly, the false FD $N \to D$ broke losslessness for Σ, so its repair moves data, while the missing FD $O \to D$ only left a key undeclared, which a declaration repairs after deployment. Hence, we ask (Q1) *Which errors of an estimate must be resolved before deployment?*

Secondly, once the designer has confirmed the three rules of Σ, the design $\{ON, OD, NC\}$ is lossless whatever else holds, but whether it preserves every rule depends on what else holds, such as $N \to D$. The sample refutes every further FD except $N \to D$ and $NC \to D$, so whether any further rule holds is one question: can two orders of the same customer have different dates? If they can, the three confirmed rules are all there is. This leads to (Q2) *Which answers certify a design, and how many are needed?*

Thirdly, an LLM that reads the attribute names can answer that a customer does not determine a date, which is right here, but its answers are wrong on many other candidates. This raises (Q3) *What can advice that may be wrong contribute to a certificate?*

Finally, designers err, and a single wrong answer can certify a wrong design. We therefore ask (Q4) *Does a certificate survive a designer who errs?*

We answer these questions for the family of synthesis algorithms, including classical 3NF synthesis and iCONF [55], that eliminate redundant FDs from the atomic closure of their input. Our main idea is to certify the design by a proof whose premises are the designer's answers and the records of the data. This certificate shows that the confirmed FDs imply exactly the true FDs, so that the guarantees of synthesis for its input become guarantees for the true FDs. It has two parts. The designer confirms a cover of the FDs that the design relies on, which makes it lossless for the true FDs. The designer also refutes every *maximal set* of the confirmed FDs, which makes it dependency-preserving and in 3NF for them as well. Such a set is maximal among the sets of attributes from which the confirmed FDs do not derive a given attribute. Refuting it is the kind of answer that Armstrong relations and design by example ask for [6, 36]: two records can agree on these attributes and differ on that one. One such answer settles every candidate for that attribute whose left-hand side lies inside the set, and the data refute some of these sets for free. On the join of the lineitem and orders tables of TPC-H, for example, a certificate needs only 49 answers among 36,883 candidates, and our algorithm finds one with 72. Since its premises are answers and records, no estimate can invalidate the certificate.

Fig. 1 shows the resulting algorithm Cert3NF. It mines the candidates and attaches to each a probability from the statistics of the data and, optionally, from the answer of an LLM, weighted by error rates learned on other schemata. We call this source of probabilities the *advisor*. Cert3NF first asks the designer to confirm the FDs of the design that the advisor believes, until that design rests on confirmed FDs alone or the designer has refuted ten of them. It then asks about the maximal sets, and synthesizes the certified design from the confirmed FDs. That is, the advisor decides the order of the questions, and the designer decides the design. A designer who may err answers every premise of the proof a second time. Our **contributions** are as follows.

(1) Conceptually, we grade the damage of an estimate by the cost of its repair (Sec. 3, Q1). Only false FDs can make a design lossy for the true FDs, and a single one does so in exactly two ways, by fabricating a key or by short-circuiting true FDs, while missing FDs always admit illegal database states. Every candidate decides the constraints that the design declares, so a certificate has to settle every candidate, and it can only afford to do so in groups.

(2) Theoretically, we show that a confirmed cover and refuted maximal sets certify a design exactly (Sec. 4, Q2). The *price* of the certificate is one answer per FD of a smallest cover and one per maximal set of the true FDs that the data do not refute, however many FDs hold by accident. No proof of the true FDs from yes and no answers is shorter.
(3) Algorithmically, we propose Cert3NF (Fig. 1, Sec. 5), in which an advisor built from the statistics of the data and, optionally, from the answers of an LLM orders the questions. Beyond one answer per refuted maximal set and per confirmed FD, a wrong advisor costs at most ten no answers and a logarithmic number of answers per FD it fails to predict, and never the correctness of the certificate (Q3). No method that does not know the true FDs avoids the logarithmic term. On data that violate true FDs in a few records, a tolerance keeps the certificate exact, since the data then refute only the FDs that fail on more than a given share of the records (Q2). For a designer who errs, every premise of the proof is answered twice, which bounds the probability of a wrong certificate (Q4).
(4) Empirically, we evaluate Cert3NF by rules fixed before any result was seen, on twelve confirmatory relations never used during development (Sec. 6). No fully automatic method that we tested, including an open LLM alone and a multi-agent normalizer, reliably obtains a good design on any of these relations. On them, a simulated designer certifies every design, without a budget, at about 2× to 2.5× the price in the median, and five to nine of the twelve within a hundred answers, depending on the ground truth. Attribute exploration, the closest classical method, needs about 3× to 5× as many answers without a budget. Every certificate is correct where the data satisfy the true FDs up to the tolerance (Q2). On synthetic joins with up to a million candidates, the answers follow the price rather than the number of candidates. Where the budget runs out, the design of Cert3NF is still lossless for the true FDs, while the interactive baselines, which accept unconfirmed FDs, return lossy designs in most such runs on the confirmatory relations. The statistics of the data alone provide most of what the advisor saves, and the two LLMs we tested add at most one certified good design and never invalidate a certificate, so the model stays optional (Q3). Answering every premise twice cuts the wrong certificates of a designer who errs independently from more than one half to about 6%. In a pre-registered study with nine people, the experts certified all their designs, in 26 minutes per session on average, while the errors of the practitioners show that a certificate is as correct as the beliefs of its designer (Q4).
(5) Practically, Cert3NF runs online at *https://cert3nf.onrender.com* for practitioners and researchers to design their own tables.

In summary, we establish the first approach to schema design that prices the certificate of a design against the unknown true FDs, lets the data refute the maximal sets that fail on them, and lets an unreliable advisor order the questions but never touch the correctness of the certificate.

Organization. Sec. 2 recalls 3NF synthesis, Sec. 3 grades errors by the cost of their repair, Sec. 4 defines the certificate and proves its price, Sec. 5 develops Cert3NF, and Sec. 6 presents our experiments. We discuss related work in Sec. 7 and conclude in Sec. 8. Appendix A contains the proofs and Appendix B the details of the experiments. Both are also in the full version of this paper, `Cert3NF-full.pdf`, in the repository of our code, online demo and results at *https://github.com/zhuoxingzhang/cert3nf*.

2 FUNDAMENTALS AND PREVIOUS WORK

We recall the notions of schema design to which the guarantees of synthesis refer, and introduce the elimination skeleton that the algorithms we study share. Ex. 1.1 illustrates them.

Dependencies. A relation schema is a finite set R of attributes. A functional dependency (FD) over R is an expression $X \to A$ with $X \subseteq R$ and $A \in R$, which a relation over R satisfies if any two of its tuples that agree on X also agree on A. For a set Σ of FDs, Σ^+ denotes the set of FDs that Σ implies [4], and $X^+_\Sigma = \{A : X \to A \in \Sigma^+\}$ the closure of X. The *atomic closure* Σ_a, which depends only on Σ^+, consists of the FDs $X \to A \in \Sigma^+$ with $A \notin X$ such that $Y \to A \notin \Sigma^+$ for every $Y \subsetneq X$. For $S \subseteq R$, the projection $\Sigma[S] = \{X \to A \in \Sigma_a : XA \subseteq S\}$ is a cover of the FDs in Σ^+ whose attributes lie within S. A key of (R, Σ) is a minimal $K \subseteq R$ with $K^+_\Sigma = R$, a superkey is a superset of a key, and an attribute is prime if it belongs to some key. A schema S is in 3NF with respect to Σ if, for every $X \to A \in \Sigma[S]$, X is a superkey of $(S, \Sigma[S])$ or A is prime for $(S, \Sigma[S])$. In Ex. 1.1, O is the only key of Orders under Σ, so $N \to C$ violates 3NF, and the projection of $\hat{\Sigma}$ onto ND is $\{N \to D\}$, while that of Σ is empty.

Designs. A design $\mathcal{D}$ over R is a decomposition of R, that is, a set of schemata whose union is R, and under a set Σ of FDs each schema $S \in \mathcal{D}$ declares the constraints $\Sigma[S]$. The design is lossless with respect to Σ if every relation over R that satisfies Σ is the natural join of its projections onto the schemata of $\mathcal{D}$, and it preserves Σ if $\bigcup_{S \in \mathcal{D}} \Sigma[S]$ implies Σ. We call $\mathcal{D}$ Σ-*valid* if it is lossless and dependency-preserving with respect to Σ and each of its schemata is in 3NF with respect to Σ. The following classical sufficient condition underlies 3NF synthesis.

Lemma 2.1 (Embedded cover [10]). *If a design $\mathcal{D}$ embeds a cover F of Σ, that is, each $X \to A \in F$ has $XA \subseteq S$ for some $S \in \mathcal{D}$, and some schema of $\mathcal{D}$ contains a key of (R, Σ), then $\mathcal{D}$ is lossless and dependency-preserving with respect to Σ.*

In Ex. 1.1, the design $\{ON, OD, NC\}$ embeds Σ and contains the key O, so it is lossless and dependency-preserving with respect to Σ.

The elimination skeleton. The synthesis algorithms we study differ only in the order in which they eliminate redundant FDs. This motivates the following definition.

Definition 2.2 (Elimination skeleton). Given Σ and an order π on Σ_a, the algorithm $\mathcal{A}_\pi$ starts with $\Sigma' = \Sigma_a$ and walks Σ_a in the order π. It removes each FD σ from Σ' that $\Sigma' \setminus \{\sigma\}$ implies, and keeps the schema XA of every FD $X \to A$ that it does not remove. The final Σ' is a non-redundant cover $G \subseteq \Sigma_a$ of Σ. The output $\mathcal{A}_\pi(\Sigma)$ consists of the maximal kept schemata, plus a minimum-size key of (R, Σ), chosen by a fixed order, if no kept schema contains a key.

That is, the skeleton keeps one schema per FD of a non-redundant cover, and the order decides which cover it keeps. Classical 3NF synthesis walks Σ_a by increasing size of the left-hand side and then lexicographically in the order of the columns, and we use its variant

that does not merge the schemata of FDs with equivalent left-hand sides. iCONF [55] parameterizes 3NF by the numbers of minimal keys and of non-key FDs of its schemata, and in the skeleton first walks the FDs whose schemata XA violate BCNF, by decreasing number of non-key FDs of XA, and then the others by decreasing number of minimal keys of XA. An FD that is walked earlier is more likely to be removed, so the order decides which schemata the result avoids. The next result shows that every order yields a valid design, and that the orders together reach every non-redundant cover within the atomic closure.

Lemma 2.3 (Skeleton guarantees). *Let π be computed from Σ_a, with ties broken by a fixed order of the attributes. Then $\mathcal{A}_\pi(\Sigma)$ depends only on Σ^+ and is Σ-valid. Over all orders, the covers G that $\mathcal{A}_\pi$ keeps are exactly the non-redundant covers of Σ contained in Σ_a.*

Hence, every member of the family inherits the guarantees of classical synthesis for the FD set it is given. What they mean for the true FDs when the given set is wrong, and how a designer can certify a design against the true FDs, have not been considered before. This is the contribution of our current work.

3 ERRORS AND THEIR DAMAGE

Synthesis runs on an estimate $\hat{\Sigma}$, while the domain imposes the true set Σ. By Lem. 2.3, the design is always $\hat{\Sigma}$-valid, so from now on, lossless, lossy, dependency-preserving and in 3NF without further qualification refer to Σ. We define the errors of $\hat{\Sigma}$ and their damage to the design, grade the damage by the cost of its repair, and show which errors a certificate must settle. We start with single errors.

Definition 3.1 (Single errors). An estimate $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ with $\sigma \notin \Sigma^+$ has the single *false FD* σ. Given a non-redundant cover G of Σ, an estimate $\hat{\Sigma} = G \setminus \{\sigma\}$ with $\sigma \in G$ has the single *missing FD* σ.

In general, the false FDs of $\hat{\Sigma}$ are those in $\hat{\Sigma}^+ \setminus \Sigma^+$, and its missing FDs those in $\Sigma^+ \setminus \hat{\Sigma}^+$. In Ex. 1.1, $N \to D$ is a single false FD and $O \to D$ a single missing FD. Errors damage a design at two levels, the declared constraints and the schemata.

Definition 3.2 (Damage). Let $\hat{\mathcal{D}}$ be synthesized from $\hat{\Sigma}$, so that each schema S declares $\hat{\Sigma}[S]$. Relative to Σ, the design *over-enforces* if some declared FD is not in Σ^+, and *under-enforces locally* if $\Sigma[S] \not\subseteq (\hat{\Sigma}[S])^+$ for some schema S. It suffers *structural damage* if it is not Σ-valid.

The two levels differ in the cost of their repair. Over- and under-enforcement are repaired by declaring $\Sigma[S]$ instead of $\hat{\Sigma}[S]$ on the existing tables, which moves no data, while structural damage requires a new decomposition and a migration of the data. In Ex. 1.1, the design $\{ON, NC, ND\}$ over-enforces $N \to D$ and suffers structural damage, since it is lossy for Σ and does not preserve $O \to D$. In contrast, the design for the missing FD $O \to D$ is Σ-valid and only under-enforces locally. The direction of an error determines which damage at the level of constraints it causes, and only false FDs can cause lossiness, as the next result shows.

Proposition 3.3 (Asymmetry). *(a) If $\hat{\Sigma} \subseteq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ is lossless with respect to Σ.*

(b) A design that preserves $\hat{\Sigma}$ over-enforces if and only if $\hat{\Sigma} \not\subseteq \Sigma^+$, and it then rejects a relation that satisfies Σ.

(c) If $\hat{\Sigma}^+ \subsetneq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ accepts a database state that is not the projection of any relation satisfying Σ.

Since synthesized designs are $\hat{\Sigma}$-valid, Prop. 3.3 applies to them: a false FD always rejects legal data, while a missing FD never causes lossiness but always admits illegal states. In Ex. 1.1, the false FD $N \to D$ rejects a second order of customer 101 on another date, and the missing FD $O \to D$ admits two dates for one order.

Every candidate decides the constraints. Damage at the level of constraints is cheap to repair, but it cannot be ruled out without settling every candidate. We call two designs *constraint-equivalent* if they have the same schemata with equivalent declared constraints.

Proposition 3.4 (Constraint equivalence). *Let $\mathcal{A}$ be an algorithm of the family, and let $\mathcal{D}_1 = \mathcal{A}(\Sigma_1)$ and $\mathcal{D}_2 = \mathcal{A}(\Sigma_2)$ preserve Σ_1 and Σ_2, respectively. If $\mathcal{D}_1$ and $\mathcal{D}_2$ are constraint-equivalent, then ${\Sigma_1}^+ = {\Sigma_2}^+$.*

Every synthesized design preserves its input by Lem. 2.3, so two candidate sets that differ in their closure synthesize designs that differ in their constraints. Hence, whether a design declares the right constraints depends on every candidate, however unlikely. In Ex. 1.1, $O \to D$ only decides whether OD declares the key O, yet a certificate must still settle it. Sec. 4 shows that the answers of a designer settle the candidates in groups, so that the certificate stays affordable.

When a false FD breaks losslessness. Lossiness is the damage that moves data, and we characterize exactly when a single false FD causes it. For the rest of this section let $\sigma = X \to a$. Let $\Sigma[\hat{\mathcal{D}}] = \bigcup_{S \in \hat{\mathcal{D}}} \Sigma[S]$ be the true FDs embedded in $\hat{\mathcal{D}}$. The following theorem shows that lossiness arises in exactly two ways.

Theorem 3.5 (Lossiness). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ have the single false FD σ, and let $\hat{\mathcal{D}} = \mathcal{A}_\pi(\hat{\Sigma})$ for any order π.*

(a) (Key loss) No schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ if and only if $a \notin S^+_\Sigma$ for every schema $S \in \hat{\mathcal{D}}$ with $S^+_{\hat{\Sigma}} = R$. In this case $\hat{\mathcal{D}}$ is lossy, and some minimal key of $(R, \hat{\Sigma})$ is not a superkey with respect to Σ.

(b) (Path loss) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ but $\hat{\mathcal{D}}$ is lossy, then $\hat{\mathcal{D}}$ does not preserve Σ, and every FD $Y \to B \in \Sigma^+$ that $\Sigma[\hat{\mathcal{D}}]$ does not imply satisfies $X \subseteq Y^+_\Sigma$.

Under *key loss*, the design relies on a key that the false FD fabricated.

Example 3.6 (Key loss). Let $R = ABCD$ and $\Sigma = \{C \to D, BC \to A\}$, with the single key BC. The false FD $D \to C$ makes BD a key as well, and both classical synthesis and iCONF walk $BC \to A$ before $BD \to A$, remove it, and synthesize $\{ABD, CD\}$, whose key schema ABD contains BD. Since $(ABD)^+_\Sigma = ABD$ does not contain C, no schema is a superkey under Σ. Hence, the design is lossy. With the attributes in reverse order, both walk $BD \to A$ first and return $\{ABC, CD\}$, which is Σ-valid, so the order of the attributes decides whether this false FD breaks losslessness. □

Under *path loss*, the design contains a true superkey, but the true FDs that would carry it to every attribute have been short-circuited. Ex. 1.1 shows path loss. The false FD $N \to D$ makes $O \to D$ redundant, so synthesis keeps the cover $\{O \to N, N \to C, N \to$

$D\}$. The schema ON is a superkey under Σ, yet $\Sigma[\hat{\mathcal{D}}] = \{O \to N, N \to C\}$ carries it only to ONC. The missing attribute D is reached only through $O \to D$, which the false FD short-circuits, as $X = N \subseteq O^+_\Sigma$. Sec. 6 shows that nine in ten accidental FDs on TPC-H make the design lossy, all by path loss. In summary, a certificate has to establish that the design is lossless, which Thm. 3.5 ties to the FDs the design relies on, and that it declares the true constraints, which Prop. 3.4 ties to every candidate. The next section shows that the answers of a designer establish both.

4 CERTIFICATES FROM ANSWERS

We now define the certificate that Cert3NF issues, and show what it costs. The designer answers one kind of question, whether an FD $Z \to A$ holds in the domain, for any $Z \subseteq R$ and $A \in R$. Two records that agree on Z and differ on A are legal exactly when $Z \to A$ does not hold, so the question can also be put as whether such records can exist, with yes and no exchanged, as with Armstrong relations [6, 36].

Setting. The *known* FDs Σ_0 are true FDs given before any question, such as the FDs of a declared key, and Σ_0 may be empty. The candidates P are the minimal FDs that hold on the data and that Σ_0 does not imply, and we assume, as FD discovery does, that the data satisfy Σ. An FD that the data violate is then not a candidate and lies outside the certificate. The designer's answers divide into the FDs Σ_y answered yes and the FDs answered no, and $H = \Sigma_0 \cup \Sigma_y$ holds everything confirmed. A candidate is *open* if H does not imply it and no answer refutes it, where a no to $Z \to B$ refutes every FD $X \to A$ with $X \subseteq Z^+_H$ and $B \in (ZA)^+_H$, since $X \to A$ and H would imply $Z \to B$. We write $\mathcal{A}$ for the synthesis algorithm in use, and call a design $\mathcal{D}$ *good* for a set Σ' of FDs if it is Σ'-valid and $|\mathcal{D}| \le |\mathcal{A}(\Sigma')|$, that is, if it has every guarantee of synthesis for Σ' and no more schemata than synthesis needs.

Soundness and completeness. A certificate has to show that the design is good for Σ, which is unknown, from H, which is known. Confirmed FDs rule out false FDs, which gives soundness, and refuted maximal sets rule out missing FDs, which gives completeness. We assume for now that the designer answers truthfully, so $H \subseteq \Sigma^+$. Hence, a design that is good for H is lossless with respect to Σ by Prop. 3.3(a), and the tables it keeps are justified by confirmed FDs alone. We call such a design *sound*. What soundness leaves open is completeness: a true FD that H does not imply may be unpreserved, violate 3NF inside a schema, or admit a smaller design. Completeness is decided by the maximal sets of attributes from which H does not derive a given attribute. This motivates the following definition.

Definition 4.1 (Maximal sets). For a set H of FDs over R and an attribute $A \in R$, the *maximal sets* $\max(H, A)$ are the maximal $M \subseteq R$ with $A \notin M^+_H$.

Thus, H implies $X \to A$ if and only if X lies within no member of $\max(H, A)$, and the members of $\max(H, A)$ are the agree sets of an Armstrong relation for H that are maximal among those not containing A [6]. In Ex. 1.1, for $H = \Sigma$, the maximal set for D is NC: two orders of the same customer, who has one city, may still differ in the date, and so may any two orders that agree on less. The significance of these sets is embodied in the following theorem.

Theorem 4.2 (Certificate). *Let $H \subseteq \Sigma^+$, and let $\mathcal{D}$ be good for H. Then $H^+ = \Sigma^+$ if and only if $M \to A \notin \Sigma^+$ for every $A \in R$ and every $M \in \max(H, A)$. In this case, $\mathcal{D}$ is good for Σ, and $H[S] = \Sigma[S]$ for every schema $S \in \mathcal{D}$, so that every schema declares the true constraints.*

Hence, the guarantees of synthesis for H become guarantees for Σ. The equivalence rests on the classical fact that the maximal sets of a set of FDs determine its closure [6], so that each maximal set becomes one question to the designer. The certificate thus consists of yes answers that justify the design and no answers that refute the maximal sets that hold on the data, and no estimate enters it. Indeed, how the candidates were estimated, and which of them an LLM judged true, can decide which questions are asked, but not whether the answers prove the design good. We illustrate the theorem on our running example.

Example 4.3 (Certificate for Orders). Consider again Ex. 1.1 with $\Sigma_0 = \emptyset$. Once the designer has confirmed $O \to N$, $O \to D$ and $N \to C$, the design $\{ON, OD, NC\}$ is good for $H = \Sigma$, so it is sound. The maximal sets are NCD for O, CD for N, D for C and NC for D. The sample refutes three of them: o_1 and o_2 agree on NCD and differ on O, o_3 and o_5 agree on CD and differ on N, and o_1 and o_3 agree on D and differ on C. Only $NC \to D$ holds on the sample, and the designer answers that two orders of the same customer can have different dates. Hence, $H^+ = \Sigma^+$, and the design is certified by three yes answers and one no answer. □

The example shows that the data can refute maximal sets without a question, and that the remaining ones are exactly those that hold on the data. For a relation r that satisfies Σ, a set $M \subseteq R$ is *held* by r if $M \in \max(\Sigma, A)$ and $M \to A$ holds on r for some attribute A. We write m_r for the number of held sets, where a set that is maximal for several attributes counts once. Every agree set of r is closed under Σ, so the FD $M \to A$ with $M \in \max(\Sigma, A)$ fails on r exactly when M is the agree set of two records of r. A set is thus held for all attributes for which it is maximal or for none. Hence, m_r counts the maximal sets of Σ that are not agree sets of r, and it is zero exactly when r is an Armstrong relation for Σ [6]. A set of answers *certifies* a design if the design is good for H and every $M \in \max(H, A)$ with $M \to A$ holding on r is refuted by a no answer, so that the design is good for Σ by Thm. 4.2. The next result makes the example precise and gives the price of a certificate.

Theorem 4.4 (Price of a certificate). *Let r satisfy Σ, and let F be a set of FDs with $(\Sigma_0 \cup F)^+ = \Sigma^+$.*

- *(a) For every $H \supseteq \Sigma_0$, a maximal set $M \in \max(H, A)$ satisfies $M \to A$ on r if and only if M contains the left-hand side of a candidate $X \to A$.*
- *(b) Yes answers to the FDs of F and, for every held set M, a no answer to one FD $M \to A$ with $M \in \max(\Sigma, A)$ that holds on r certify every design that is good for Σ, with $|F| + m_r$ answers in all.*
- *(c) Every set of answers that certifies a design contains at least m_r no answers, and its yes answers Σ_y satisfy $(\Sigma_0 \cup \Sigma_y)^+ = \Sigma^+$.*
- *(d) A set of answers certifies a design that is good for Σ if and only if Σ^+ is the only closed set of FDs that holds on r, contains Σ_0 and the yes answers, and contains no FD answered no.*

We call $|F| + m_r$ for a smallest F the *price* of a certificate, and Thm. 4.4 shows that it is set by the domain and the data, not by the

number of candidates. Indeed, the yes answers are the cover that a designer would write down anyway, and the no answers are one per maximal set that the data do not refute, where one no settles at once every candidate that leads from inside the set to an attribute outside it. By (d), any answers that leave Σ^+ as the only closed set of FDs that agrees with them and holds on the data form a certificate, and by (c), these answers contain a set F and one no answer per held set. Hence, the price is the length of a shortest proof of the true FDs from yes and no answers. A smallest F is a minimum Horn base, which is NP-hard to find [19], but integer programming finds it in practice (Appendix B). FDs with sets of attributes on the right would have a smallest cover in polynomial time [34], but one yes would then confirm several FDs at once, while each of our questions asks about one attribute.

Part (a) also shows why the certificate refutes maximal sets rather than candidates. A true FD $Y \to A$ whose left-hand side is not minimal on the data is not a candidate, since some $X \subsetneq Y$ determines A on the data by accident. Refuting the candidate $X \to A$ would leave $Y \to A$ unsettled. In contrast, while H does not imply $Y \to A$, some $M \in \max(H, A)$ contains Y, so $M \to A$ holds on the data by (a) and is true, and the designer's yes to it reveals a missing FD inside M. The maximal sets can be exponentially many [6], but the price counts only the held ones. Since maximal sets are usually large, the question for M is best put by the attributes outside M: must two records that agree on everything but these also agree on A? Thm. 4.4 assumes that the right questions are asked, and the next section shows how to find them.

5 THE ALGORITHM CERT3NF

Cert3NF has to find the cover and the held sets of Σ without knowing Σ, within a budget of B answers, which we set to 100. It works in two phases, shown in Fig. 1. The discovery phase confirms the cover of the design that an *advisor* believes, and the certification phase refutes the maximal sets of the confirmed FDs. The advisor is any source of a probability per candidate, and ours is built from the statistics of the data and, optionally, from the answers of an LLM. It only orders the questions, and we bound what a wrong advisor can cost.

The advisor. An estimator assigns each candidate $c \in P$ a probability ρ_c from statistics of the data, such as the size of its left-hand side and the numbers of distinct values, by a logistic model trained on other schemata. Estimators trained elsewhere rank candidates well but are rarely calibrated on a new schema, so the advisor uses $\text{logit}^{-1}(\kappa \cdot \text{logit}\, \rho_c + \lambda)$ instead, with a prior on (κ, λ) learned on other schemata. An LLM may be asked whether c holds, in one prompt that names the attributes, states Σ_0 and shows a few rows. Its answer enters as evidence through its likelihood ratio, $(1-\beta)/\alpha$ for a yes and $\beta/(1-\alpha)$ for a no, when the model answers no to a true candidate with rate β and yes to a false one with rate α. The rates differ widely between schemata, from almost none to over four fifths on the development relations of Sec. 6, so each schema has its own pair (α, β) under a prior learned on other schemata. The designer's answers about candidates update the posterior over (κ, λ) and (α, β), and the probability ρ'_c of the advisor is the posterior mean. The model is asked only where its answer can change the next question, that is, about the candidates that a yes would make believed and about the likeliest ones inside a maximal set before a guess.

Discovery. The advisor *believes* the open candidates with $\rho'_c \geq 1/2$. These FDs and the confirmed FDs $H = \Sigma_0 \cup \Sigma_y$ form its estimate $\hat{\Sigma}$, and the *target design* is $\mathcal{A}(\hat{\Sigma})$. Let $\hat{G}$ be a non-redundant subset of the believed candidates that generates $\hat{\Sigma}$ with H, that is, the unconfirmed FDs on which the target design relies. Cert3NF asks the designer about the most likely FD of $\hat{G}$. A yes puts the FD into H, a no refutes it and every candidate that would imply it with H (Sec. 4), and the advisor, $\hat{\Sigma}$, the target design and $\hat{G}$ are recomputed. The phase ends when $\hat{G}$ is empty, so that $\hat{\Sigma}^+ = H^+$ and the target design is good for H and hence sound, or when b answers were no, with $b = 10$. Every yes answer puts a true FD into H, so an advisor that insists on false FDs costs at most b no answers before the maximal sets take over. An advisor that believes more true FDs than a smallest cover needs costs yes answers instead, which Prop. 5.2 counts.

Certification. From then on Cert3NF asks about the maximal sets of H. By Thm. 4.4(a), a pair (M, A) with $M \in \max(H, A)$ needs a question only if M contains the left-hand side of a candidate of A; otherwise two records of r refute it without a question. Among the pairs that need one, Cert3NF prefers those with an open candidate inside M, and asks about the pair for which $M \to A$ is most likely, that is, for which $1 - \prod_c (1 - \rho'_c)$ over the open candidates c of A inside M is largest. A no refutes at once every candidate $X \to A'$ with $X \subseteq M$ and $A' \notin M$. A yes means that an FD $X \to A$ with $X \subseteq M$ is missing from H, and Cert3NF finds a minimal one. It guesses first, asking about the likeliest open candidate inside M while its probability given $M \to A$ is at least $\max(0.2, 1/g)$, for at most $g = \lceil \log_2 |M| \rceil + 1$ guesses. A wrong guess is not wasted, since it refutes a candidate. If the guesses miss, it asks about subsets of M by QuickXplain [25], with the attributes of M ordered by how much probability the open candidates place on them, and finds a left-hand side of k attributes with $O(k(1 + \log(|M|/k)))$ further answers. The FD found joins H, and the maximal sets of H are recomputed from the minimal keys of $H \cup \{A \to B : B \in R\}$ [33] by transversals. The phase ends when no pair needs a question, that is, when every maximal set of H that holds on r is refuted. By Thm. 4.2, $H^+ = \Sigma^+$, and Cert3NF returns $\mathcal{A}(H)$ with its certificate: the confirmed FDs and the refuted pairs, each refuted by a no answer or by two records of r.

In Ex. 4.3, the statistics make O look like a key and the advisor doubts $N \to D$, so it believes $O \to N$, $O \to D$ and $N \to C$. The designer confirms them in three answers, and one more refutes the only maximal set that holds on the data, NC for D. If the advisor also doubted $O \to D$, the designer would confirm $O \to N$ and $N \to C$, answer yes to the maximal set ONC for D, confirm the guess $O \to D$ inside it and refute NC, in five answers. The no answers to maximal sets stay within the price.

Proposition 5.1 (Refuted maximal sets). *Let r satisfy Σ and the designer answer truthfully. Every no answer to a maximal set in the certification phase falls on a held set, and no held set receives two. Hence, the phase receives at most m_r no answers to maximal sets.*

The next result adds the cap b of the discovery phase and the questions that find each FD, and so counts all answers.

Proposition 5.2 (Number of answers). *Let r satisfy Σ and the designer answer truthfully, and let $J(k,n) \in O(k(1+\log(n/k)))$ be the number of answers that QuickXplain needs to find a left-hand side of k attributes among n. Every yes answer of the discovery phase and every FD found in the certification phase is a different FD of Σ^+ that Σ_0 does not imply, with a left-hand side that is minimal under Σ. If d FDs enter H in all, Cert3NF asks at most*

$$b + m_r + d + \sum_{X \to A \text{ found by a yes to } (M,A)} \big(g + J(|X|, |M|)\big)$$

questions.

Hence, beyond the m_r no answers to maximal sets and one answer per FD, an advisor costs at most b no answers in discovery and g wrong guesses per FD. The FDs that a yes reveals are found at a logarithmic cost, whatever the advisor says. On our relations, d is at most 2.5× the size of a smallest cover in the median, although it can be exponentially larger in general (Appendix A). The logarithmic term is the cost of not knowing Σ, and no method avoids it.

Proposition 5.3 (Lower bound). *For all $1 \le k < m$, let $R = MA$ with $|M| = m$ and $\Sigma_0 = \emptyset$. There is a relation r over R on which every $\Sigma = \{X \to A\}$ with $X \subseteq M$ and $|X| = k$ holds and has the price $k+1$, while every method that certifies a design for each of them needs at least $\log_2 \binom{m}{k} \ge k \log_2(m/k)$ answers on one of them.*

Hence, no method certifies every design within a constant factor of the price. Without an advisor, Cert3NF needs at most $1 + k + J(k,m)$ answers on these relations, which is within a constant factor of the best method in the worst case, as every method also pays the price. An advisor that believes exactly $X \to A$ brings the cost down to the price.

Partial certificates. If the budget runs out, Cert3NF certifies what it can. A schema S of $\mathcal{A}(H)$ is certified on its own when every maximal set of the projection $H[S]$ is refuted, which leaves only the preservation of FDs across the schemata open.

Proposition 5.4 (Partial certificate). *Let $H \subseteq \Sigma^+$, let $S \subseteq R$, and let $T \to A \notin \Sigma^+$ for every $A \in S$ and every $T \in \max(H[S], A)$. Then Σ^+ and H^+ contain the same FDs over S, and a schema S of $\mathcal{A}(H)$ is in 3NF with respect to Σ and has $H[S] = \Sigma[S]$.*

Without a certificate, Cert3NF returns the sound design $\mathcal{A}(H)$ with its certified schemata. The design is lossless, its certified schemata already declare the true constraints, and a later session continues from its answers.

A designer who errs. The certificate of Thm. 4.2 is a proof, and a wrong answer is a false premise. The *premises* are a non-redundant subset of the yes answers that generates H with Σ_0, and a subset of the no answers that refutes every maximal set of H that holds on r. If all of them are right, so is the certificate. In its *careful mode*, Cert3NF therefore has the designer answer every premise a second time before it certifies, and a third time when the two answers disagree. The net answer of a question is the majority of its answers, and a question with one answer of each kind counts as unanswered. The set H holds the net yes answers only, so the designer is asked once more about the FD that QuickXplain finds. No question is asked more than three times, and the certificate is issued once every premise has a majority of at least two answers (Appendix A gives the rule for conflicting answers).

Proposition 5.5 (Wrong certificates). *Let r satisfy Σ, let every answer be wrong independently with probability ε, and let Q be the expected number of different questions that Cert3NF asks in its careful mode. Then Cert3NF issues a wrong certificate with probability at most $Q(3\varepsilon^2 - 2\varepsilon^3)$, which is below $300\varepsilon^2$ within a budget of 100 answers.*

That is, if a designer errs on one answer in fifty, at least seven runs in eight end without a wrong certificate, on any data that satisfy Σ. The bound holds as well when every question has its own error rate ε_q, drawn independently from one distribution, with $3\mathrm{E}[\varepsilon_q^2] - 2\mathrm{E}[\varepsilon_q^3]$ in place of $3\varepsilon^2 - 2\varepsilon^3$. A designer who repeats an error has $\varepsilon_q \in \{0, 1\}$, for which this term equals ε, the error rate of a single answer.

Data that violate true FDs. The certificate also assumes that the data satisfy Σ, while real data violate true FDs in a few records. The data then refute such an FD without a question, so the designer never sees it. We say that $M \to A$ holds on r *up to* θ if at most a share θ of the records has to be removed for $M \to A$ to hold. This share is the measure g_3 of Kivinen and Mannila [27], which discovery of approximate FDs uses [23]. With a *tolerance* θ, Cert3NF asks the designer about every maximal set $M \in \max(H, A)$ for which $M \to A$ holds on r up to θ, and lets the data refute the others. The next result shows that the certificate then stays exact on data that satisfy the true FDs up to θ.

Proposition 5.6 (Tolerance). *Let every FD of Σ^+ hold on r up to θ, and read "holds on r" as "holds on r up to θ" and the candidates as the minimal FDs that hold on r up to θ and that Σ_0 does not imply. Then the data refute no FD of Σ^+, a set of answers that certifies a design proves it good for Σ, and Thm. 4.4(a)–(c) hold.*

Every maximal set that the data no longer refute costs a question, so a tolerance trades questions for robustness.

The posterior certificate. Instead of the maximal sets, a posterior over closed sets of candidates can certify a design by asking about the candidates of largest posterior marginal and accepting a sound design once it is good for a share $1 - \delta$ of the closed sets sampled from the posterior. We use this *posterior certificate* as a baseline. It rests on the error rates of the model and on the candidates. Sec. 6 shows that it is wrong where the model refutes a true FD or a true FD is not minimal on the data, while the certificate of Thm. 4.2 cannot be.

6 EXPERIMENTS

Our experiments answer the empirical side of the four questions of the introduction, which build one argument. Single errors damage most synthesized designs, so designs must be certified (Q1). No automatic method reliably obtains a good design, so a designer pays the price of a certificate (Q2). An advisor lowers this cost (Q3), and certificates survive the slips of a designer (Q4).

6.1 Set up

Data. Random schemata and the eight tables of TPC-H [44] measure the damage of single errors (Q1). The certificates are evaluated on 36 relations with data and documented FDs, with 7 to 25 attributes and 200 to 2.7 million rows (Tab. 8 in Appendix B). Cert3NF was designed on the 24 *development* relations (23 with answers of the model), and its prior and the error rates of the model are learned

on them. The 12 *confirmatory* relations were chosen from their documentation alone by rules fixed in advance, and never opened during development. Four of them are benchmarks whose schema a model may have memorized (class A), and eight are real data sets with a data dictionary (class B).

Ground truth. The ground truth of a relation consists of the FDs that its publisher documents, in a *full* version that includes the rules a reader has to infer and a *sure* version of the explicit ones. Together with the declared key as Σ_0, it plays the designer, and for (Q4) we inject wrong answers. A design is *good* if it is good for the documented FDs and the declared key (Sec. 4), and a certificate is *wrong* if it rests on a false FD or misses a documented FD that holds on the data. On five confirmatory relations, listed before any run and marked † in the tables, the data violate a documented FD of the full truth, which then lies outside every certificate.

Methods. We compare Cert3NF with two ablations: *without LLM*, whose advisor is the statistics alone, and *without advisor*, which runs the certification phase alone, with the maximal sets in a fixed order and QuickXplain in place of the guesses. The tables show the *posterior certificate* of Sec. 5 with the model, and the EC^2 *certificate*, which asks about the candidates whose answers separate the most pairs of posterior samples with different designs [18]. The fully automatic methods ask no designer: the mined FDs taken as true (*mined FDs*), the yes answers of the model (*trust*), a per-FD classifier on the statistics and the answer (*classifier*), the posterior alone (*posterior alone*), and the rules that the model states for the relation (*rules*). The model also designs the relation itself, when asked directly (*LLM design*) and as a multi-agent normalizer with a verifier after MARS [28] (*normalizer*). As a reference, *every candidate* lets the designer judge every candidate without a budget.

Protocol. The candidates are all minimal FDs of the data, mined by our exact implementation in the style of HyFD [39], and the LLM is Qwen3.5-9B. Cert3NF is deterministic, and every method with a random component runs with three seeds. Before any confirmatory result was seen, we fixed five rules by which the claims are judged, per truth on all relations and on each class (Appendix B). They state that every certificate is correct where the data satisfy the documented FDs (C1), that the model certifies more relations or saves answers under both truths (C2), that the advisor certifies at least as many relations and costs at most b answers plus one per maximal set answered yes on nine in ten relations (C3), that the runs certified within the budget pay at most 2× the price in the median (C4), and that Cert3NF has at least as many good designs as the posterior certificate (C5). Before running the EC^2 certificate, we fixed that Cert3NF has at least as many good designs and certificates (C6).

6.2 Q1: Which Errors Damage the Design?

We first ask which errors damage a synthesized design, and how often.

Random schemata. Tab. 1 shows that one error causes structural damage to about two thirds of the designs for $n \leq 6$, and to up to 87.5% for $n = 7$, whichever algorithm synthesizes them. Indeed, every order walks the same wrong atomic closure with the same wrong keys and only selects among its covers. Hence, robustness to errors has to come from the estimate or from the designer, not from the choice of the synthesis algorithm. As Prop. 3.3 predicts (first five rows of Tab. 1), every false FD over-enforces and half of them make the design lossy, while missing FDs do neither and break preservation or 3NF instead.

Table 1: Damage by a Single Error on Random Schemata with $n \leq 6$ Attributes (in %, $n = 7$ in Parentheses, – No Lossy Design)

	false FD		missing FD	
measure	classical	iCONF	classical	iCONF
over-enforcing	100 (100)	100 (100)	0 (0)	0 (0)
locally under-enforcing	0 (0)	0 (0)	73.1 (64.0)	73.0 (64.0)
lossy	51.1 (62.8)	50.8 (62.6)	0 (0)	0 (0)
not preserving	41.5 (28.0)	41.5 (27.1)	44.5 (59.2)	44.6 (59.2)
not in 3NF	10.4 (10.1)	8.5 (7.8)	30.8 (43.8)	30.7 (43.8)
structural damage	68.6 (73.5)	68.2 (72.8)	68.0 (87.5)	68.1 (87.5)
key loss, of lossy	63.6 (85.5)	64.1 (86.2)	–	–

Table 2: Damage by a Single Accidental FD on TPC-H (in %, iCONF in Parentheses Where It Differs)

table	$\lvert R\rvert$	accidental FDs	lossy	not in 3NF	structural damage
region	3	2	100	0	100
nation	4	3	100	0	100
partsupp	5	16	100	0	100
supplier	7	21	85.7	14.3 (4.8)	90.5
customer	8	34	94.1	0	94.1
orders	8	44	100	0	100
part	9	79	86.1	7.6 (3.8)	89.9
lineitem	16	3,670	89.3	0.6 (0.4)	89.8
all	–	3,869	89.5	0.8 (0.5)	90.0

TPC-H. Tab. 2 shows that accidental FDs damage the structure even more often than random false FDs: 90.0% of them cause structural damage, against 68.6% on random schemata. Indeed, in every table the primary key K determines each other attribute directly, so the left-hand side X of every accidental FD $X \to a$ lies in $K^+_\Sigma = R$, which is the condition of Thm. 3.5(b) for short-circuiting $K \to a$. Unless a true non-key FD also reaches a or leads from a into X, synthesis then stores a with X instead of with K, and the join on X, which is not a superkey under Σ, is lossy, as in Ex. 1.1. Where X becomes a key as well, the order of the walk decides whether $K \to a$ or $X \to a$ is removed, as in Ex. 3.6, and Tab. 2 breaks ties by the order of the columns. The two mechanisms of Thm. 3.5 thus dominate on different data: fabricated keys cause most lossy designs on random schemata (last row of Tab. 1), and path loss causes every lossy design on TPC-H. In summary, a single false or missing FD damages the structure of most synthesized designs, so a wrong estimate costs a migration of the data, not just a worse design. Hence, both kinds of error have to be resolved before deployment, and certification has to settle almost every plausible accidental FD.

Table 3: Answers until the Certificate on the Confirmatory Relations, with True Candidates and Price (Thm. 4.4) under the Full and the Sure Truth; in Parentheses the Answers without a Budget, Where They Exceed 100, – No Certificate, † a Design That Misses a Documented FD the Data Violate; Posterior: Certified Runs of Three and Their Median Answers; $\theta = 0.1$: Cert3NF on the Unrepaired Data with This Tolerance

		full truth							sure truth						
relation	cand.	true	price	Cert3NF	w/o LLM	w/o advisor	posterior	$\theta = 0.1$	true	price	Cert3NF	w/o LLM	w/o advisor	posterior	$\theta = 0.1$
auctionmark (A)	4,243	36	33	47	45	(114)	–	47	15	47	58	58	(125)	2 (93.5)	58
seats (A)	4,640	292	54	(124†)	(135†)	(345†)	–	(122)	19	40	69	62	(122)	–	69
tatp (A)	909	28	38	(188)	60	81	–	(190)	28	38	76	63	81	–	78
simplewiki (A)	139	36	44	(122†)	(137†)	(121†)	–	(139)	30	40	(140)	(149)	80	2 (100)	(172)
chembl	385	204	54	(238†)	(381†)	(674†)	–	(106)	43	29	75	58	(156)	1 (100)	96
musicbrainz	85	71	35	71†	(105†)	(170†)	3† (80)	88	61	34	46	70	(137)	3 (73)	59
pluto	1,572	202	108	(322†)	(449†)	(488†)	–	(375†)	46	42	(292)	(283)	(111)	–	(345)
fhrs	1,488	25	30	(154)	(169)	70	–	(168)	25	30	(131)	(167)	70	–	(180)
geonames	2,900	12	43	85	84	(144)	–	92	12	43	78	80	(144)	–	85
eia-860	1,097	14	35	51	51	100	–	63	14	35	49	49	100	–	61
stats19	1,670	42	33	(108)	(104)	79	3 (60)	(127)	42	33	98	(116)	79	3 (55)	(117)
ghcn-daily	351	13	28	37	43	80	3 (98)	(123)	13	28	41	41	80	3 (80)	56
within 100				5/12	5/12	5/12	9/36	4/12			9/12	8/12	6/12	14/36	8/12

Table 4: Good Designs, by the Median of Three Seeds, and Macro F1 of the Selected FDs, Averaged over the Seeds; EC^2 Ran on the Confirmatory Relations Only

	good designs				F1			
	dev.		conf.		dev.		conf.	
method	full	sure	full	sure	full	sure	full	sure
mined FDs	0/23	0/23	0/12	0/12	.32	.20	.22	.14
trust	0/23	0/23	0/12	0/12	.38	.27	.25	.17
classifier	0/23	0/23	0/12	0/12	.38	.55	.52	.53
posterior alone	0/23	0/23	0/12	0/12	.44	.33	.39	.35
rules	0/23	0/23	0/12	0/12	.44	.36	.27	.23
LLM design	0/23	0/23	0/12	0/12	.20	.27	.17	.17
normalizer	1/23	3/23	0/12	0/12	.39	.46	.29	.37
every candidate	16/23	18/23	7/12	12/12	1.00	1.00	1.00	1.00
posterior certificate	14/23	16/23	4/12	8/12	.94	.97	.74	.89
EC^2 certificate	–	–	2/12	6/12	–	–	.65	.74
Cert3NF	18/23	21/23	5/12	9/12	.93	.95	.84	.94

6.3 Q2: What Does a Certified Design Cost?

Since almost every accidental FD must be settled, we ask whether an automatic method can do so, and otherwise what a certificate costs a designer.

Automatic methods. Tab. 4 shows that, by the median of its seeds, no fully automatic method obtains a good design on a confirmatory relation, and only the normalizer does on development relations, one of 23 under the full truth and three under the sure one. Indeed, a good design has to preserve every true FD and avoid the damage of false ones, and each method fails at least one of the two. The mined FDs often contain thousands of accidental ones, the model refutes up to four fifths of the true candidates of a relation or omits attributes and rules in the designs it writes, and the normalizer infers only part of the rules. Graded by the damage of Sec. 3, the designs of these methods are lossless on at most five (full) and three (sure) of the twelve confirmatory relations, and none of the methods avoids structural damage on any relation in the median of its seeds. Even a designer who judges every candidate obtains a good design only where every true FD is a minimal FD of the data and the data satisfy Σ. Indeed, on six development relations a documented FD is not minimal on the data, so no candidate expresses it, and on five confirmatory relations the data violate one. Hence, the designer is not optional and the candidates are not enough, and the question becomes what a designer has to answer.

Certificates within the budget. Tab. 3 shows that Cert3NF certifies the designs of five of the twelve confirmatory relations under the full truth and of nine under the sure truth within a hundred answers. Every certificate is correct where the data satisfy the documented FDs (C1 holds). The certified runs pay a median of 1.46× (full) and 1.73× (sure) the price of Thm. 4.4 (C4 holds). On the development relations (Tab. 9), Cert3NF certifies 18 of 23 (full) and 21 of 23 (sure) at a median of 1.58× and 1.65× the price.

Interactive baselines. The posterior certificate trusts the candidates and the model, and errs where a true FD is not minimal on the data or the model refutes a true FD: on the development relations, it is wrong in 8 of 40 certified runs under the full truth and in 12 of 43 under the sure truth (Tab. 9), while Cert3NF is never wrong. On the confirmatory relations, no certificate of any method is wrong where the data satisfy the documented FDs, since every documented FD that holds there is minimal on the data, and the methods differ in their designs. At the end of the budget, Cert3NF has at least as many good designs as the posterior certificate on every class and truth (C5 holds). Where both certify a confirmatory relation, Cert3NF needs 29 (full) and 28 (sure) fewer answers than the posterior certificate in the median over its certified runs. Indeed, a no to a maximal set settles every candidate that leads from inside it to an attribute outside it, while the posterior certificate asks about them one by one.

Choosing the questions by EC^2 does not raise the good designs (C6 holds): the EC^2 certificate certifies 11 (full) and 12 (sure) of

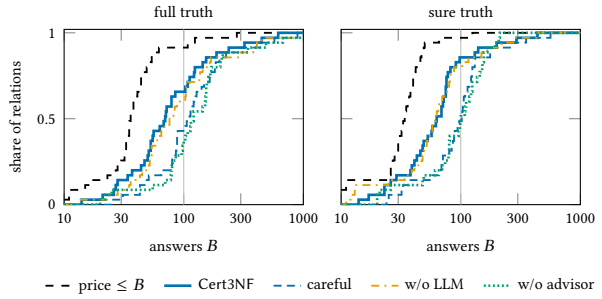


Figure 2: Share of the 35 Relations (23 Development, 12 Confirmatory) Certified within B Answers, and Share Whose Price, a Bound for Every Method, Is at Most B

Table 5: Answers until the Certificate on Synthetic Relations without a Budget (Medians over Three Seeds; Share: of the Other Attributes on the Entity Side; Ratio: Answers to Price)

					Cert3NF w/o LLM		w/o advisor
attr.	share	cand.	true	price	answers	ratio	answers
10	0.6	78	5	17	21	1.24	38
20	0.6	16,294	19	42	64	1.60	154
30	0.2	554,220	9	59	71	1.20	113
30	0.4	806,080	20	63	78	1.24	153
30	0.6	145,161	40	61	105	1.79	256
30	0.8	284,217	53	61	112	1.75	268

36 runs, against 9 and 14 for the posterior certificate, but has fewer good designs than both the posterior certificate and Cert3NF (Tab. 4). Indeed, EC^2 stops asking about a candidate once all samples agree on it, so 24 of its 25 uncertified runs under the full truth rely on FDs that the designer has not confirmed.

Designs without a certificate. Where the budget runs out before the certificate, the design of Cert3NF is still lossless, since it rests on confirmed FDs alone. In contrast, the posterior and EC^2 certificates rely on unconfirmed FDs where they stop, and on the confirmatory relations only 9 of 27 and 3 of 25 of their uncertified designs are lossless under the full truth. On these relations, the uncertified designs of Cert3NF are also certified in parts: 205 of their 220 schemata under the full truth and 65 of 69 under the sure truth are certified on their own (Prop. 5.4).

Without a budget. Fig. 2 shows that only the budget stops the uncertified runs: without it, Cert3NF certifies every relation (parentheses in Tab. 3 and Tab. 9), nine in ten within 250 answers, which take about two hours at the pace of the experts in our study (Sec. 6.5). The certified runs then pay a median of 1.96× and 1.84× the price over all relations, and 2.53× and 1.91× on the confirmatory ones. Only seven (full) and five (sure) of the 35 relations exceed 3× the price, where many maximal sets are answered yes and finding the FD behind each yes costs several answers.

Attribute exploration. The closest classical method has an expert answer whether an implication holds or construct a counterexample [15, 16]. It determines Σ as exactly as a certificate does, but at several times the answers (Appendix B). With the smallest counterexamples, it does not finish within two hours on three (full) and four (sure) confirmatory relations. With the largest, which are maximal sets, it poses a median of 28 implications, fewer than the price. However, confirming an implication takes a yes per attribute of its conclusion, and refuting it takes an answer per attribute by which the counterexample grows. Its implications thus amount to a median of 8.6× (full) and 10.2× (sure) the price in yes and no answers, and to 2.8× and 4.6× those of Cert3NF without a budget. Indeed, the price bounds proofs from yes and no answers, and most answers of exploration grow the counterexamples, whereas the designer of Cert3NF only judges the maximal sets that the confirmed FDs determine.

Scalability. Tab. 5 shows that the domain and the data, not the number of candidates, set the cost of a certificate. On synthetic snowflake joins whose true FDs hold by construction (Appendix B), the candidates grow from 78 at 10 attributes to about a million on the largest run at 30. Yet the price stays below 73 on every run, and Cert3NF without the LLM certifies every design correctly at a median of 1.5× the price. Indeed, the accidental candidates multiply with the attributes, but the maximal sets settle them in groups. Hence, the answers follow the price (Spearman's rank correlation 0.73 over the 18 relations) and the true FDs (0.87) rather than the candidates (0.41), which confirms the criterion we fixed before generating the data. Without an advisor, Cert3NF needs about 2× as many answers (last column), so the statistics alone halve the cost where the names of the attributes carry no meaning. Cert3NF takes less than a second of computation per answer in the median, also with a million candidates, and 3.4 seconds at most. Beyond 30 attributes, mining the candidates and holding them in memory set the limit (Appendix B).

Violated FDs. A certificate assumes that the data satisfy Σ, and the tolerance of Prop. 5.6 keeps it exact on data that violate true FDs in a few records. Before all runs, a greedy repair removed records until the sure FDs hold (Appendix B), and the columns $\theta = 0.1$ of Tab. 3 run on the data before the repair. Of the five † relations, only pluto keeps its †, since its documented rule fails on 16% of the records, and every certificate within the budget is correct. Without a tolerance, every certificate on the four relations that the repair changed misses a documented FD under either truth. The tolerance also saves answers, as on fhrs, where it halves them under the sure truth, since the designer no longer confirms many FDs with larger left-hand sides in place of each FD that the data violate. Hence, $\theta = 0.1$, the largest of the three tolerances fixed before these runs, is a good default, which costs about twelve answers in the median without a budget on the repaired data.

In summary, the cost of a certificate is set by the cover and the held sets of a schema rather than by its candidates, and the certificate is never wrong where the data satisfy the true FDs up to the tolerance.

6.4 Q3: What Does an Advisor Buy?

Since the certificate does not depend on the advisor, we ask what the advisor contributes and what a wrong one costs.

The model. The model adds few certificates on real data. On the eight real data sets of class B (Tab. 3), Cert3NF certifies four relations against three (full) and six against five (sure) with the

model rather than without it, with a median difference of zero answers where both certify, so C2 holds on this class. The added certificate under the full truth is the † design of musicbrainz, and the one under the sure truth is good. Over all twelve relations, Cert3NF certifies nine against eight under the sure truth and five either way under the full truth, where a model that misleads the guesses, as on tatp, costs answers but not correctness. Since C2 asks for a gain under both truths, it does not hold over all relations, and the model stays an optional part of the advisor.

A second model. A stronger model does not change this, since two models err on the same candidates. With gpt-oss-20b, which errs on 27% of the confirmatory candidates while Qwen errs on 34%, Cert3NF certifies seven relations under the full truth and eight under the sure truth, against five and eight without a model. Its two added certificates are † designs, so the number of good designs stays the same under both truths, and no certificate is wrong. Indeed, where one model errs, the odds that the other errs are 4× higher, so a stronger model lowers the error rate but leaves the hard candidates to the designer.

The advisor. Without an advisor, the certification phase alone certifies 13 of the 35 relations within the budget under the full truth and 18 under the sure truth, against 23 and 30 with it, of which the statistics alone reach 21 and 28 (Fig. 2 at $B = 100$). On the confirmatory relations, the advisor gains three certificates and loses three within the budget under the full truth, and gains five and loses two under the sure truth, and without a budget the phase alone needs 54 (full) and 45 (sure) more answers in the median. Indeed, QuickXplain has to find every revealed FD one attribute at a time, where the advisor guesses it in one. Prop. 5.2 bounds the damage of a wrong advisor, and it stays small in practice. Within the budget, Cert3NF needs at most b more answers than the phase without an advisor, plus one per maximal set answered yes, on 11 of the 12 confirmatory relations and on all development relations (C3 holds over all relations and on class A). Without a budget, this holds on 10 (full) and 9 (sure) of the 12. The exception within the budget, by which C3 misses class B, is fhrs, where the advisor puts first the maximal sets most likely to hold and pays for the FDs their yes answers reveal.

6.5 Q4: Does a Certificate Survive a Designer Who Errs?

The designer of (Q2) and (Q3) never errs. We now let the simulated designer answer wrongly with probability $\varepsilon \in \{0.02, 0.05\}$, independently per answer and with three seeds, and compare the careful mode of Cert3NF with a trusting mode that accepts all answers.

Independent errors. Tab. 6 shows that a certificate that trusts every answer breaks under a few wrong ones, while the careful mode keeps the wrong certificates within the bound of Prop. 5.5. On the development relations, trusting the designer makes 19 of 44 certificates wrong at $\varepsilon = 0.02$ and 31 of 38 at $\varepsilon = 0.05$ under the full truth. On these relations, answering every premise twice reduces the wrong certificates to one or two per truth and error rate, about 3% of the certificates at $\varepsilon = 0.02$ and at most 10% at $\varepsilon = 0.05$. They are at most 3% of the runs, within the 4% and 23% that Prop. 5.5 gives for the 34 and 32 different questions of the median certified run.

Table 6: Certificates and Wrong Certificates (✗) of a Designer Who Errs with Probability ε, over Relations and Three Seeds (69 Development and 36 Confirmatory Runs per Cell); Repeated: the Designer Repeats a Wrong Answer; ϕ: Correlation of Two Answers to a Question

		development				confirmatory			
		full		sure		full		sure	
ε	mode	cert.	✗	cert.	✗	cert.	✗	cert.	✗
0.02	trusting	44	19	51	21	11	7	14	8
0.02	careful	31	1	32	1	4	1	4	1
0.05	trusting	38	31	35	28	6	4	8	8
0.05	careful	19	1	21	2	3	0	1	0
0.05, repeated	careful	24	18	23	17	3	1	2	2
0.02, $\phi = 0.25$	careful	25	3	32	3	5	1	7	1
0.02, $\phi = 0.5$	careful	26	12	27	8	4	0	5	0
0.05, $\phi = 0.25$	careful	20	12	19	8	0	0	1	1
0.05, $\phi = 0.5$	careful	22	12	22	11	1	0	1	0

Table 7: Nine Designers, Three per Condition (E: Experts; P, U: Practitioners with and without Documentation); Block: Fixed Questions about Candidates and Maximal Sets; $\hat{\phi}$: Correlation of Two Answers to a Question

errors (%)	E	P	U	sessions	E	P	U
block candidates	0	3	12	answers per session	55	46	45
block maximal sets	0	8	17	minutes per session	26	35	36
in the sessions	1	10	22	certified, of six	6	3	1
don't know (%)	0	3	7	wrong, full truth	0	1	0
$\hat{\phi}$	0.1	0.7	0.8	wrong, sure truth	1	1	1

Second answers. The careful mode pays for this correctness in answers: with a designer who never errs, it asks every premise once more and nothing else, so that a certificate costs the answers of the trusting mode plus its premises, between 1.2× and 1.9× as many and 1.54× in the median (Fig. 2). Within the budget, it certifies 15 (full) and 17 (sure) of the 35 relations, three of them confirmatory, and without a budget every relation, nine in ten within 300 answers. However, with errors only 2 of its 12 confirmatory certificates are wrong, against 27 of 39 when the designer is trusted, and 7 of 115 over all relations and both truths, against 126 of 207.

Correlated errors. Asking again catches the part of the errors that is independent, and no repetition can catch a designer who repeats a wrong answer (row "repeated" of Tab. 6). Between independent and repeated errors, every question has its own error rate, drawn with mean ε so that two answers to it err together with correlation ϕ (last rows of Tab. 6). Pooled over all relations and both truths, the share of wrong certificates at $\varepsilon = 0.02$ grows from 6% at $\phi = 0$ to 12% at $\phi = 0.25$ and 32% at $\phi = 0.5$, still below the 46% of the trusting mode.

Human designers. Tab. 7 reports a pre-registered study with nine people, in which the experts certified their designs in one session and the practitioners mostly did not (Appendix B). Three

database experts, who studied the documentation before each session, and six practitioners, half of whom read it for ten minutes in the session, each worked with the careful mode of Cert3NF on two of ssb-customer, tpcds-core and abs, within a hundred answers or forty minutes per session. The experts certified all six of their designs, with 55 answers in 26 minutes per session on average. They erred on 1% of their answers and on none of a fixed set of questions about candidates and maximal sets, and none of their certificates is wrong under the full truth. Their one certificate that is wrong under the sure truth confirms an inferred FD, which only the full truth contains. The practitioners erred on 22% of their answers and certified one of six designs without the documentation, and erred on 10% and certified three of six with ten minutes of it. Two answers of theirs to the same question err together with correlation $\hat{\phi} = 0.8$ and 0.7, so their errors are beliefs rather than slips. Hence, the careful mode catches slips, and no proof from answers can catch false beliefs: a certificate is as correct as the beliefs of its designer.

Summary. A single error damages the structure of most synthesized designs (Q1). No fully automatic method reliably obtains a good design on a confirmatory relation, while a simulated designer asked about the maximal sets certifies every design without a budget, at about 2× to 2.5× the price in the median on the confirmatory relations and 1.5× on synthetic relations with up to a million candidates (Q2). Where a budget runs out, the design of Cert3NF is still lossless, while on the confirmatory relations most designs of the interactive baselines are lossy. An advisor, mostly by the statistics of the data, lifts the relations certified within the budget from 13 to 23 (full) and from 18 to 30 (sure) of 35, and from six to nine on the confirmatory ones under the sure truth, while five stay five under the full truth. A wrong advisor never costs the correctness of the certificate (Q3). Answering every premise twice catches most independent errors of a designer, experts certified all their designs in our study, and a certificate is as correct as the beliefs of its designer (Q4).

7 RELATED WORK

Schema design. 3NF synthesis [7, 10] and its recent optimizations for integrity maintenance and update efficiency [31, 49, 53–55] assume that the given FDs are correct. Wang et al. normalize a mediated schema with probabilistic FDs estimated from web sources [47], and Link and Prade scale 3NF and BCNF to degrees of certainty of FDs [30]. In both, the uncertainty is part of the input and the design is judged against it, while we judge it against the unknown true FDs.

Dependency discovery. FD discovery algorithms return all FDs that hold on an instance [23, 38, 39], and Dep-Miner derives them from the agree sets and maximal sets of the data [32]. Wei and Link rank the discovered FDs by the redundancy they cause [48], and Wan and Han discover only the k FDs that cause the most [46]. Others find reliable FDs by chance correction, statistical tests, or models of noise and missing values [8, 20, 35, 52]. Normalize chooses among the discovered FDs to decompose a relation into BCNF [40]. These approaches improve the estimate, but leave errors that only a designer can settle.

Confirming dependencies. Armstrong relations satisfy exactly the FDs implied by a given set [4, 6], and Mannila and Räihä let designers confirm FDs by example [36]. Langeveldt and Link show in a user study that they help designers recognize the meaningful FDs [29], and De Marchi and Petit build them from the records of an existing database [37]. The maximal sets we ask about are the agree sets of such relations and the characteristic models of the Horn theory that the FDs form [26]. Our certification phase thus asks the questions of design by example, but only about the held sets, which the data cannot refute, and Thm. 4.4 shows that no proof needs fewer. Formal concept analysis treats FDs as implications over the pairs of records [5], and its attribute exploration asks an expert whether the implications of a minimal basis hold, where the expert refutes an implication by a counterexample [15, 16]. Seeded with the agree sets, it determines Σ as a certificate does, but its expert constructs the counterexamples and its questions follow the basis rather than the design (Sec. 6.3). Learning such a theory [2, 14] or a minimal cover of FDs [22] needs, besides membership or entailment queries, equivalence queries, which return counterexamples, or closure queries, which return a whole closed set [3], whereas our designer only answers yes or no. By Thm. 4.4(d), our answers form a teaching set [17, 21] for Σ^+ among the closed sets of FDs that hold on the data, and by part (c) the price is its smallest size, which, unlike the teaching dimension of a class of concepts, depends on the data. Guided data repair [50] and UGuide [43] ask users about repairs and FDs within a budget to find the errors that FDs detect.

LLMs as oracle or designer. LLMs predict data correlations from column names [45], and have been asked instead of people as the oracle that decides which statistically strong FDs are meaningful [51], generates ontological FDs for data cleaning [9], or detects foreign keys [13, 42]. They have also been made the designer that normalizes schemata [24], where inferring the FDs is the main bottleneck [28], or that drafts domain models from requirements [11], also with a human who answers questions about their most uncertain parts [41]. In both roles nothing bounds the risk that the result is wrong, and in our experiments neither role reliably obtains a good design on a confirmatory relation. Our work is the first to price the certificate of a design against the unknown true FDs, and to let the data and an unreliable advisor lower its cost but not its correctness.

8 CONCLUSION AND FUTURE WORK

Starting from the observation that estimated FD sets are wrong, we have shown how a designer certifies a schema design against the true FDs at a price set by the domain and the data rather than by the candidates. A confirmed cover and refuted maximal sets settle the candidates in groups, and a smallest cover with the held sets is a shortest proof from yes and no answers. When the data form an Armstrong relation for the true FDs, the certificate is a confirmed smallest cover, the input of classical design.

Future work will address i) designers who answer by example rather than by yes or no, as with Armstrong relations [36], ii) the certification of several relations linked by foreign keys, iii) certificates for BCNF, and iv) stronger models as advisors. Our results show that the answers of a designer who knows the domain follow the length of the proof, and that advice that may be wrong can only change the order of the questions.

REFERENCES

[1] Alfred V. Aho, Catriel Beeri, and Jeffrey D. Ullman. 1979. The Theory of Joins in Relational Databases. *ACM Trans. Database Syst.* 4, 3 (1979), 297–314. https://doi.org/10.1145/320083.320091

[2] Dana Angluin, Michael Frazier, and Leonard Pitt. 1992. Learning Conjunctions of Horn Clauses. *Machine Learning* 9, 2-3 (1992), 147–164. https://doi.org/10.1007/BF00992675

[3] Marta Arias, José L. Balcázar, and Cristina Tîrnăucă. 2017. Learning Definite Horn Formulas from Closure Queries. *Theor. Comput. Sci.* 658 (2017), 346–356. https://doi.org/10.1016/j.tcs.2015.12.019

[4] William Ward Armstrong. 1974. Dependency Structures of Data Base Relationships. In *Information Processing, Proceedings of the 6th IFIP Congress 1974, Stockholm, Sweden, August 5-10, 1974*. North-Holland, 580–583.

[5] Jaume Baixeries, Mehdi Kaytoue, and Amedeo Napoli. 2014. Characterizing Functional Dependencies in Formal Concept Analysis with Pattern Structures. *Ann. Math. Artif. Intell.* 72, 1-2 (2014), 129–149. https://doi.org/10.1007/s10472-014-9400-3

[6] Catriel Beeri, Martin Dowd, Ronald Fagin, and Richard Statman. 1984. On the Structure of Armstrong Relations for Functional Dependencies. *J. ACM* 31, 1 (1984), 30–46. https://doi.org/10.1145/2422.322414

[7] Philip A. Bernstein. 1976. Synthesizing Third Normal Form Relations from Functional Dependencies. *ACM Trans. Database Syst.* 1, 4 (1976), 277–298. https://doi.org/10.1145/320493.320489

[8] Laure Berti-Équille, Hazar Harmouch, Felix Naumann, Noël Novelli, and Saravanan Thirumuruganathan. 2018. Discovery of Genuine Functional Dependencies from Relational Data with Missing Values. *Proc. VLDB Endow.* 11, 8 (2018), 880–892. https://doi.org/10.14778/3204028.3204032

[9] Fabian Biester, Mohamed Abdelaal, and Daniel Del Gaudio. 2025. LLMClean: Context-Aware Tabular Data Cleaning via LLM-Generated OFDs. In *New Trends in Database and Information Systems - ADBIS 2024 Short Papers, Workshops, Doctoral Consortium and Tutorials, Bayonne, France, August 28-31, 2024, Proceedings (Communications in Computer and Information Science)*, Vol. 2186. Springer, 68–78. https://doi.org/10.1007/978-3-031-70421-5_7

[10] Joachim Biskup, Umeshwar Dayal, and Philip A. Bernstein. 1979. Synthesizing Independent Database Schemas. In *Proceedings of the 1979 ACM SIGMOD International Conference on Management of Data*. ACM, 143–151. https://doi.org/10.1145/582095.582118

[11] Kua Chen, Yujing Yang, Boqi Chen, José Antonio Hernández López, Gunter Mussbacher, and Dániel Varró. 2023. Automated Domain Modeling with Large Language Models: A Comparative Study. In *26th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems, MODELS 2023, Västerås, Sweden, October 1-6, 2023*. IEEE, 162–172. https://doi.org/10.1109/MODELS58315.2023.00037

[12] E. F. Codd. 1970. A Relational Model of Data for Large Shared Data Banks. *Commun. ACM* 13, 6 (1970), 377–387. https://doi.org/10.1145/362384.362685

[13] Till Döhmen, Radu Geacu, Madelon Hulsebos, and Sebastian Schelter. 2024. SchemaPile: A Large Collection of Relational Database Schemas. *Proc. ACM Manag. Data* 2, 3 (2024), 172:1–172:25. https://doi.org/10.1145/3654975

[14] Michael Frazier and Leonard Pitt. 1993. Learning From Entailment: An Application to Propositional Horn Sentences. In *Proceedings of the Tenth International Conference on Machine Learning (ICML)*. Morgan Kaufmann, 120–127. https://doi.org/10.1016/B978-1-55860-307-3.50022-8

[15] Bernhard Ganter. 2010. Two Basic Algorithms in Concept Analysis. In *Formal Concept Analysis, 8th International Conference, ICFCA 2010, Proceedings*. Springer, 312–340. https://doi.org/10.1007/978-3-642-11928-6_22

[16] Bernhard Ganter and Sergei A. Obiedkov. 2016. *Conceptual Exploration*. Springer. https://doi.org/10.1007/978-3-662-49291-8

[17] Sally A. Goldman and Michael J. Kearns. 1995. On the Complexity of Teaching. *J. Comput. Syst. Sci.* 50, 1 (1995), 20–31. https://doi.org/10.1006/JCSS.1995.1003

[18] Daniel Golovin, Andreas Krause, and Debajyoti Ray. 2010. Near-Optimal Bayesian Active Learning with Noisy Observations. In *Advances in Neural Information Processing Systems 23 (NIPS)*. Curran Associates, Inc., 766–774.

[19] Peter L. Hammer and Alexander Kogan. 1993. Optimal Compression of Propositional Horn Knowledge Bases: Complexity and Approximation. *Artif. Intell.* 64, 1 (1993), 131–145. https://doi.org/10.1016/0004-3702(93)90062-G

[20] Ziyan Han, Yeye He, Shuyuan Kang, Min Xie, Weiwei Cui, Song Ge, Haidong Zhang, Dongmei Zhang, Surajit Chaudhuri, Rui Mao, and Jianbin Qin. 2026. AutoRelate: A Unified Approach to Discovering Reliable Functional Relationships Leveraging Statistical Tests. *CoRR* abs/2606.07060 (2026). https://doi.org/10.48550/arXiv.2606.07060

[21] Lisa Hellerstein, Krishnan Pillaipakkamnatt, Vijay Raghavan, and Dawn Wilkins. 1996. How Many Queries Are Needed to Learn? *J. ACM* 43, 5 (1996), 840–862. https://doi.org/10.1145/234752.234755

[22] Montserrat Hermo and Víctor Lavín. 1999. Learning Minimal Covers of Functional Dependencies with Queries. In *Algorithmic Learning Theory, 10th International Conference, ALT '99, Tokyo, Japan, December 6-8, 1999, Proceedings (Lecture Notes in Computer Science)*, Vol. 1720. Springer, 291–300. https://doi.org/10.1007/3-540-46769-6_24

[23] Ykä Huhtala, Juha Kärkkäinen, Pasi Porkka, and Hannu Toivonen. 1999. TANE: An Efficient Algorithm for Discovering Functional and Approximate Dependencies. *Comput. J.* 42, 2 (1999), 100–111. https://doi.org/10.1093/comjnl/42.2.100

[24] Eunjae Jo, Nakyung Lee, and Gyuyeong Kim. 2025. Database Normalization via Dual-LLM Self-Refinement. *CoRR* abs/2508.17693 (2025). https://doi.org/10.48550/arXiv.2508.17693

[25] Ulrich Junker. 2004. QUICKXPLAIN: Preferred Explanations and Relaxations for Over-Constrained Problems. In *Proceedings of the Nineteenth National Conference on Artificial Intelligence (AAAI)*. AAAI Press / The MIT Press, 167–172.

[26] Roni Khardon, Heikki Mannila, and Dan Roth. 1999. Reasoning with Examples: Propositional Formulae and Database Dependencies. *Acta Informatica* 36, 4 (1999), 267–286. https://doi.org/10.1007/s002360050161

[27] Jyrki Kivinen and Heikki Mannila. 1995. Approximate Inference of Functional Dependencies from Relations. *Theor. Comput. Sci.* 149, 1 (1995), 129–149. https://doi.org/10.1016/0304-3975(95)00028-U

[28] Dong-Jae Koh, Huisu Kim, Seonghwan Yoon, Lasse M. Jantsch, Chun-Hee Lee, Seonghyeon Lee, and Young-Kyoon Suh. 2026. Can LLMs Normalize Databases? A Benchmark and Multi-Agent Framework for Schema Normalization. *CoRR* abs/2609.11141 (2026). https://doi.org/10.48550/arXiv.2609.11141

[29] Warren-Dean Langeveldt and Sebastian Link. 2010. Empirical evidence for the usefulness of Armstrong relations in the acquisition of meaningful functional dependencies. *Inf. Syst.* 35, 3 (2010), 352–374. https://doi.org/10.1016/J.IS.2009.11.002

[30] Sebastian Link and Henri Prade. 2019. Relational database schema design for uncertain data. *Inf. Syst.* 84 (2019), 88–110. https://doi.org/10.1016/j.is.2019.04.003

[31] Sebastian Link and Ziheng Wei. 2021. Logical Schema Design that Quantifies Update Inefficiency and Join Efficiency. In *SIGMOD '21: International Conference on Management of Data, Virtual Event, China, June 20-25, 2021*. ACM, 1169–1181. https://doi.org/10.1145/3448016.3459238

[32] Stéphane Lopes, Jean-Marc Petit, and Lotfi Lakhal. 2000. Efficient Discovery of Functional Dependencies and Armstrong Relations. In *Advances in Database Technology - EDBT 2000, 7th International Conference on Extending Database Technology (Lecture Notes in Computer Science)*, Vol. 1777. Springer, 350–364. https://doi.org/10.1007/3-540-46439-5_24

[33] Claudio L. Lucchesi and Sylvia L. Osborn. 1978. Candidate Keys for Relations. *J. Comput. Syst. Sci.* 17, 2 (1978), 270–279. https://doi.org/10.1016/0022-0000(78)90009-0

[34] David Maier. 1983. *The Theory of Relational Databases*. Computer Science Press.

[35] Panagiotis Mandros, Mario Boley, and Jilles Vreeken. 2017. Discovering Reliable Approximate Functional Dependencies. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 355–363. https://doi.org/10.1145/3097983.3098062

[36] Heikki Mannila and Kari-Jouko Räihä. 1986. Design by Example: An Application of Armstrong Relations. *J. Comput. Syst. Sci.* 33, 2 (1986), 126–141. https://doi.org/10.1016/0022-0000(86)90015-2

[37] Fabien De Marchi and Jean-Marc Petit. 2007. Semantic sampling of existing databases through informative Armstrong databases. *Inf. Syst.* 32, 3 (2007), 446–457. https://doi.org/10.1016/J.IS.2005.12.007

[38] Thorsten Papenbrock, Jens Ehrlich, Jannik Marten, Tommy Neubert, Jan-Peer Rudolph, Martin Schönberg, Jakob Zwiener, and Felix Naumann. 2015. Functional Dependency Discovery: An Experimental Evaluation of Seven Algorithms. *Proc. VLDB Endow.* 8, 10 (2015), 1082–1093. https://doi.org/10.14778/2794367.2794377

[39] Thorsten Papenbrock and Felix Naumann. 2016. A Hybrid Approach to Functional Dependency Discovery. In *Proceedings of the 2016 International Conference on Management of Data (SIGMOD)*. ACM, 821–833. https://doi.org/10.1145/2882903.2915203

[40] Thorsten Papenbrock and Felix Naumann. 2017. Data-driven Schema Normalization. In *Proceedings of the 20th International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 342–353. https://doi.org/10.5441/002/edbt.2017.31

[41] Jonathan Silva Mercado, Qin Ma, Jordi Cabot, Pierre Kelsen, and Henderik A. Proper. 2026. Towards Human-in-the-Loop LLM-Enabled Domain Modeling. In *Conceptual Modeling - 44th International Conference, ER 2025, Poitiers, France, October 20-23, 2025, Proceedings (Lecture Notes in Computer Science)*, Vol. 16189. Springer, 127–145. https://doi.org/10.1007/978-3-032-08623-5_7

[42] Zijian Tang, Ying Zhang, Sibo Cai, and Ruoxuan Wang. 2026. LLM-FK: Multi-Agent LLM Reasoning for Foreign Key Detection in Large-Scale Complex Databases. In *Findings of the Association for Computational Linguistics, ACL 2026, San Diego, California, United States, July 2-7, 2026*. Association for Computational Linguistics, 33201–33228. https://doi.org/10.18653/v1/2026.findings-acl.1660

[43] Saravanan Thirumuruganathan, Laure Berti-Équille, Mourad Ouzzani, Jorge-Arnulfo Quiané-Ruiz, and Nan Tang. 2017. UGuide: User-Guided Discovery of FD-Detectable Errors. In *Proceedings of the 2017 International Conference on Management of Data (SIGMOD)*. ACM, 1385–1397. https://doi.org/10.1145/3035918.3064024

[44] Transaction Processing Performance Council. 2022. TPC Benchmark H Standard Specification, Revision 3.0.1. https://www.tpc.org/tpch/

[45] Immanuel Trummer. 2023. Can Large Language Models Predict Data Correlations from Column Names? *Proc. VLDB Endow.* 16, 13 (2023), 4310–4323. https://doi.org/10.14778/3625054.3625066

[46] Xiaolong Wan and Xixian Han. 2026. Redundancy-Driven Top-k Functional Dependency Discovery. *CoRR* abs/2601.10130 (2026). https://doi.org/10.48550/arXiv.2601.10130

[47] Daisy Zhe Wang, Xin Luna Dong, Anish Das Sarma, Michael J. Franklin, and Alon Y. Halevy. 2009. Functional Dependency Generation and Applications in Pay-As-You-Go Data Integration Systems. In *12th International Workshop on the Web and Databases (WebDB)*.

[48] Ziheng Wei and Sebastian Link. 2023. Towards the efficient discovery of meaningful functional dependencies. *Inf. Syst.* 116 (2023), 102224. https://doi.org/10.1016/j.is.2023.102224

[49] Ziheng Wei and Sebastian Link. 2025. The Bounded Cardinality Normal Form for the Logical Design of Relational Database Schemata. *ACM Trans. Database Syst.* 50, 4 (2025), 15:1–15:45. https://doi.org/10.1145/3744897

[50] Mohamed Yakout, Ahmed K. Elmagarmid, Jennifer Neville, Mourad Ouzzani, and Ihab F. Ilyas. 2011. Guided Data Repair. *Proc. VLDB Endow.* 4, 5 (2011), 279–289. https://doi.org/10.14778/1952376.1952378

[51] Shuo Zhang, Zezhou Huang, and Eugene Wu. 2025. Data Cleaning Using Large Language Models. In *41st IEEE International Conference on Data Engineering, ICDE 2025 - Workshops, Hong Kong, May 19-23, 2025*. IEEE, 28–32. https://doi.org/10.1109/ICDEW67478.2025.00008

[52] Yunjia Zhang, Zhihan Guo, and Theodoros Rekatsinas. 2020. A Statistical Perspective on Discovering Functional Dependencies in Noisy Data. In *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*. ACM, 861–876. https://doi.org/10.1145/3318464.3389749

[53] Zhuoxing Zhang, Wu Chen, and Sebastian Link. 2023. Composite Object Normal Forms: Parameterizing Boyce-Codd Normal Form by the Number of Minimal Keys. *Proc. ACM Manag. Data* 1, 1 (2023), 13:1–13:25. https://doi.org/10.1145/3588693

[54] Zhuoxing Zhang and Sebastian Link. 2024. Mixed Covers of Keys and Functional Dependencies for Maintaining the Integrity of Data under Updates. *Proc. VLDB Endow.* 17, 7 (2024), 1578–1590. https://doi.org/10.14778/3654621.3654626

[55] Zhuoxing Zhang and Sebastian Link. 2025. Synthesizing Third Normal Form Schemata that Minimize Integrity Maintenance and Update Overheads: Parameterizing 3NF by the Numbers of Minimal Keys and Functional Dependencies. *Proc. ACM Manag. Data* 3, 3 (2025), 225:1–225:25. https://doi.org/10.1145/3725362

A PROOFS

We prove the results in the order of the paper. Losslessness is decided by the chase of the tableau of $\mathcal{D}$ [1, 34], which has one row per schema S, with the distinguished symbol of column A for every $A \in S$ and fresh symbols elsewhere. The chase with Σ equates the symbols in column A of rows that agree on X, for $X \to A \in \Sigma$, and prefers the distinguished symbol. The design $\mathcal{D}$ is lossless if and only if some row ends with distinguished symbols only. Several proofs reuse the argument of Lem. 2.1. It chases the row of the schema S that contains a key along a derivation of $S^+_\Sigma = R$ by the FDs of F. When the derivation applies $X \to A$, embedded in S', the rows of S and S' agree on X in distinguished symbols, so the row of S receives the distinguished symbol in column A.

Proof of Lem. 2.3. The first claim holds because Σ_a, the keys and the fixed choices depend only on Σ^+. The union of the output is R: an attribute in no kept schema occurs in no FD of G and so lies in every key, hence no kept schema contains a key, and the added key contains the attribute. The output also embeds G and contains a key, so it is lossless and dependency-preserving by Lem. 2.1. Each kept schema XA is in 3NF: X is a key of XA by the minimality of X, and a non-trivial FD $Z \to B$ within XA either has $B = A$, so that $Z \subseteq X$ and hence $Z = X$, or has $B \in X$, which is prime. The key schema K carries no non-trivial FD, since $K \setminus \{B\}$ would otherwise be a superkey of R. For the last claim, the kept covers are non-redundant subsets of Σ_a by construction. Conversely, a non-redundant cover $G \subseteq \Sigma_a$ is kept by an order that walks $\Sigma_a \setminus G$ first: each of these FDs is implied by $G \subseteq \Sigma'$ and removed, after which $\Sigma' = G$ and no FD of G is redundant. □

Proof of Prop. 3.3. (a) Every relation that satisfies Σ satisfies $\hat{\Sigma}$ and is therefore the join of its projections.

(b) If some declared FD is not in Σ^+, it is in $\hat{\Sigma}^+$, so $\hat{\Sigma} \not\subseteq \Sigma^+$. Conversely, if every declared FD is in Σ^+, preservation gives $\hat{\Sigma} \subseteq (\bigcup_S \hat{\Sigma}[S])^+ \subseteq \Sigma^+$. For the rejection, let $Y \to B \in \hat{\Sigma}[S] \setminus \Sigma^+$ and take two tuples that agree exactly on Y^+_Σ. They satisfy Σ, since the set on which they agree is closed, but they violate $Y \to B$, and so does their projection onto $S \supseteq YB$.

(c) Let $\sigma \in \Sigma^+ \setminus \hat{\Sigma}^+$ and let r be a two-tuple relation that satisfies $\hat{\Sigma}$ and violates σ. The projections of r satisfy the declared constraints. If they were the projections of some s that satisfies Σ, then s would satisfy $\hat{\Sigma}$, and losslessness with respect to $\hat{\Sigma}$ would give $s = \bowtie_S s[S] = \bowtie_S r[S] = r$, which violates σ. □

Proof of Prop. 3.4. We have

$$\Sigma_1 \subseteq (\textstyle\bigcup_S \Sigma_1[S])^+ = (\bigcup_S \Sigma_2[S])^+ \subseteq {\Sigma_2}^+,$$

and $\Sigma_2 \subseteq {\Sigma_1}^+$ holds symmetrically. □

The proof of Thm. 3.5 rests on two lemmas, which describe what one more FD does to closures and to the chase.

Lemma A.1 (One more FD). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$. Then $Z^+_{\hat{\Sigma}} = Z^+_\Sigma$ if $X \not\subseteq Z^+_\Sigma$, and $Z^+_{\hat{\Sigma}} = (Za)^+_\Sigma$ otherwise.*

Proof. Let Z' denote the right-hand side, which is contained in $Z^+_{\hat{\Sigma}}$ since $\hat{\Sigma} \supseteq \Sigma$ and $a \in Z^+_{\hat{\Sigma}}$ whenever $X \subseteq Z^+_\Sigma$. Conversely, Z' contains Z, is closed under Σ, and is closed under σ, because $X \not\subseteq Z'$ in the first case and $a \in Z'$ in the second, so $Z^+_{\hat{\Sigma}} \subseteq Z'$. □

Let $\hat{\Sigma}$ and $\hat{\mathcal{D}}$ be as in Thm. 3.5, let $\hat{G}$ be the cover that $\mathcal{A}_\pi$ keeps on $\hat{\Sigma}$, and let W_S be the set of columns in which the row of schema S holds distinguished symbols after the chase of the tableau of $\hat{\mathcal{D}}$ with Σ.

Lemma A.2 (Rows and superkeys). *(a) If $X \not\subseteq S^+_\Sigma$ for a schema $S \in \hat{\mathcal{D}}$, then $W_S = S^+_\Sigma = S^+_{\hat{\Sigma}}$.*

(b) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to $\Sigma[\hat{\mathcal{D}}]$, then $\hat{\mathcal{D}}$ is lossless with respect to Σ. If $\hat{\mathcal{D}}$ is lossless with respect to Σ, then some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ. The two conditions coincide when $\hat{\mathcal{D}}$ preserves Σ.

Proof. (a) For the upper bound $W_S \subseteq S^+_\Sigma$, let t and s be two tuples that agree exactly on S^+_Σ, so that $\{t, s\}$ satisfies Σ. Mapping the row of S to s, every other row to t, and each distinguished symbol to the value of t in its column is a homomorphism from the tableau into $\{t, s\}$, because s and t agree on S. Every chase step equates two symbols that this homomorphism already maps to the same value, so the chased tableau maps into $\{t, s\}$ as well. A column in which the row of S is distinguished is therefore one in which s and t agree, that is, a column in S^+_Σ. For the lower bound, $S^+_{\hat{\Sigma}} = S^+_\Sigma$ by Lem. A.1, and $S^+_{\hat{\Sigma}}$ is derived from S by the FDs of $\hat{G}$. Every FD $Z \to C$ used in this derivation has $Z \subseteq S^+_\Sigma$, so $X \not\subseteq Z^+_\Sigma$, and it is in Σ^+ by Lem. A.1, and as an FD of $\hat{G}$ it is embedded in some schema of $\hat{\mathcal{D}}$. Following the derivation as in the proof of Lem. 2.1, the row of S receives the distinguished symbol in every column of $S^+_{\hat{\Sigma}}$.

(b) The first statement follows from the argument of Lem. 2.1, since every FD of $\Sigma[\hat{\mathcal{D}}]$ is true and embedded in a schema of $\hat{\mathcal{D}}$. For the second, a row of distinguished symbols only belongs to a schema S with $R = W_S \subseteq S^+_\Sigma$ by the upper bound in (a). If $\hat{\mathcal{D}}$ preserves Σ, then $(\Sigma[\hat{\mathcal{D}}])^+ = \Sigma^+$, and the two conditions coincide. □

Proof of Thm. 3.5. (a) Let $S^+_{\hat{\Sigma}} = R$. If $a \in S^+_\Sigma$, then $S^+_\Sigma = S^+_{\hat{\Sigma}} = R$ by Lem. A.1, whether or not $X \subseteq S^+_\Sigma$, and conversely $S^+_\Sigma = R$ contains a. A schema with $S^+_{\hat{\Sigma}} \neq R$ is not a superkey under Σ either, since $S^+_\Sigma \subseteq S^+_{\hat{\Sigma}}$. This proves the equivalence, and lossiness follows from Lem. A.2(b). Finally, $\mathcal{A}_\pi$ ensures that some schema contains a minimal key K of $(R, \hat{\Sigma})$, and if $K^+_\Sigma = R$, that schema is a superkey under Σ.

(b) The first statement is the contrapositive of the last sentence of Lem. A.2(b). For the second, suppose that $Y \to B \in \Sigma^+$ is not implied by $\Sigma[\hat{\mathcal{D}}]$ while $X \not\subseteq Y^+_\Sigma$. Then $Y^+_{\hat{\Sigma}} = Y^+_\Sigma$ by Lem. A.1, and as in the proof of Lem. A.2(a) the derivation of B from Y by $\hat{G}$ only uses FDs of Σ^+ that are embedded in schemata S' of $\hat{\mathcal{D}}$. Each of them is implied by $\Sigma[S']$, so $\Sigma[\hat{\mathcal{D}}]$ implies $Y \to B$, a contradiction. □

Proof of Thm. 4.2. Suppose that $M \to A \notin \Sigma^+$ for every A and every $M \in \max(H, A)$, and let $X \to A \in \Sigma^+$. If H did not imply $X \to A$, then $X \subseteq M$ for some $M \in \max(H, A)$, and $M \to A$ would follow from $X \to A$ by augmentation, a contradiction. Hence, $\Sigma^+ \subseteq H^+$, and $H \subseteq \Sigma^+$ gives equality. Conversely, if $H^+ = \Sigma^+$ and $M \in \max(H, A)$, then $M \to A \notin H^+ = \Sigma^+$. Finally, validity depends only on the closure of the FD set by its definition, and so does $\mathcal{A}$ by Lem. 2.3, so a design that is good for H is good for Σ. The atomic closure also depends only on the closure, so $H[S] = \Sigma[S]$ for every schema S. □

Proof of Thm. 4.4. (a) If M contains the left-hand side of a candidate $X \to A$, then $M \to A$ holds on r by augmentation, since every candidate holds on r. Conversely, if $M \to A$ holds on r, let $X \subseteq M$ be minimal with $X \to A$ holding on r. Since $H \supseteq \Sigma_0$ does not imply $M \to A$, the FD $X \to A$ is not implied by Σ_0, so it is a minimal FD of r outside ${\Sigma_0}^+$, that is, a candidate.

(b) After the yes answers, $H = \Sigma_0 \cup F$, so $H^+ = \Sigma^+$ and $\max(H, A) = \max(\Sigma, A)$ for every A. A maximal set on which $M \to A$ fails on r is refuted by two tuples of r, which satisfies Σ, so $M \to A \notin \Sigma^+$ without a question, and the remaining ones are the held sets. Let M be held, let $M \to A$ be its no answer, and let $M \in \max(\Sigma, A')$ with $M \to A'$ holding on r. Then $M = M_H^+$, since M is closed under Σ, and $A \in (MA')_H^+$, since $(MA')_\Sigma^+$ is a closed set that contains M and $A' \notin M$, so it contains A by the maximality of M for A. Hence, the no answer to $M \to A$ refutes $M \to A'$, every maximal set of H is false, and Thm. 4.2 certifies every design that is good for H, that is, for Σ.

(c) Consider a set of answers that certifies a design. Every maximal set of H is false, so $(\Sigma_0 \cup \Sigma_y)^+ = H^+ = \Sigma^+$ by Thm. 4.2. The held sets are then the $M \in \max(H, A)$ for which $M \to A$ holds on r, each refuted by a no answer to some $Z \to B$ with $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$. If $A \in Z_H^+$, then $B \in (ZA)_H^+ = Z_H^+$ and $Z \to B \in H^+ = \Sigma^+$, which contradicts the truthful no. Hence, Z_H^+ is a closed set that contains M and not A, and $M = Z_H^+$ by the maximality of M. Thus a no answer to $Z \to B$ refutes the maximal sets of the one held set Z_H^+, and a certificate has at least one no answer per held set.

(d) Let a set of answers certify a design that is good for Σ, so that $H^+ = \Sigma^+$ as in (c), and let Σ' be a closed set of FDs that holds on r, contains H, and contains no FD answered no. Then $\Sigma^+ \subseteq \Sigma'$. If $X \to A \in \Sigma' \setminus \Sigma^+$, then $X \subseteq M$ for some $M \in \max(\Sigma, A) = \max(H, A)$, and $M \to A \in \Sigma'$ holds on r, so a no answer to some $Z \to B$ refutes it. That is, $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so $Z \to B \in \Sigma'$, a contradiction. Hence, $\Sigma' = \Sigma^+$. Conversely, let Σ^+ be the only such set. The closure of H is one, since r satisfies Σ and $H^+ \subseteq \Sigma^+$ contains no FD answered no, so $H^+ = \Sigma^+$. For $M \in \max(H, A)$ with $M \to A$ holding on r, the closure of $H \cup \{M \to A\}$ holds on r and differs from Σ^+, so it contains an FD $Z \to B$ answered no. Since $B \notin Z_H^+$, this requires $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so the no answer refutes $M \to A$. Hence, every maximal set of H that holds on r is refuted, and the answers certify every design that is good for H, that is, for Σ. □

Proof of Prop. 5.1. Let the phase ask about $M \to A$ with $M \in \max(H, A)$ and receive no. Since $H \subseteq \Sigma^+$ and $M \to A \notin \Sigma^+$, $M \subseteq M'$ for some $M' \in \max(\Sigma, A)$; M' is closed under Σ, hence under H, and does not contain A, so $M = M'$ by the maximality of M in $\max(H, A)$. The phase asks only about maximal sets that contain the left-hand side of a candidate, so $M \to A$ holds on r by Thm. 4.4(a), and M is held. For every attribute $A' \notin M$, $(MA')_H^+$ is a closed set that strictly contains M, so it contains A by the maximality of M, and it only grows with H. Hence, the no answer to $M \to A$ refutes $M \to A'$ under every later H, so no later question concerns M. □

Proof of Prop. 5.2. Every candidate holds on r and is minimal there, so a true candidate has a minimal left-hand side under Σ as well, since a smaller true left-hand side would hold on r. A yes answer of the discovery phase or a guess that hits therefore puts into H a true candidate with a minimal left-hand side that H did not imply, and QuickXplain returns a minimal $X \subseteq M$ with $X \to A \in \Sigma^+$, which H did not imply since M is closed under H and does not contain A. Once $X \to A$ is in H, no later maximal set for A contains X, so the same FD is not found twice, and no later question concerns a candidate in H; hence the d FDs are different, and Σ_0 implies none of them. The questions are of four kinds. The discovery phase receives at most b no answers and one yes per FD that enters H. The certification phase receives at most m_r no answers to maximal sets by Prop. 5.1, one yes per FD that it finds, at most g guesses after that yes, and $J(|X|, |M|)$ questions of QuickXplain if the guesses miss. Summing gives the bound. The count d can be exponential in the size of a smallest cover: if Σ consists of $A_i \to B_i$ and $B_i \to A_i$ for $i = 1, \ldots, n$ and of $A_1 \cdots A_n \to C$, a run that settles C before the equivalences confirms one FD for each of the 2^n ways to pick A_i or B_i. □

Proof of Prop. 5.3. Let Σ' consist of the FDs $Y \to A$ with $Y \subseteq M$ and $|Y| = k$, and let r be an Armstrong relation for Σ' [6], on which an FD holds exactly when Σ' implies it. For a k-subset X of M, $\Sigma = \{X \to A\}$ holds on r, and a smallest F has one FD. The maximal sets of Σ are $M \setminus \{x\}$ for A with $x \in X$, and $R \setminus \{B\}$ for $B \in M$. Since $m - 1 \geq k$, Σ' implies $M \setminus \{x\} \to A$, while Σ' implies no FD $R \setminus \{B\} \to B$. Hence, the held sets are the k sets $M \setminus \{x\}$, and the price is $k + 1$. A method chooses each question from the earlier answers, so its runs form a binary tree, and two sets $X \neq X'$ whose runs end in the same leaf receive the same answers. These answers are truthful for both $\{X \to A\}$ and $\{X' \to A\}$, which hold on r and have different closures, so by Thm. 4.4(d) they certify a design for at most one of them. Hence, the tree has at least $\binom{m}{k}$ leaves and a depth of at least $\log_2 \binom{m}{k} \geq k \log_2(m/k)$. □

Proof of Prop. 5.4. For $T \subseteq S$, $T_{H[S]}^+ = T_H^+ \cap S$, since $H[S]$ is a cover of the FDs of H^+ within S. Hence, every $T \subseteq S$ with $A \notin T_H^+$ lies within a member of $\max(H[S], A)$. Let $X \to A \in \Sigma^+$ with $XA \subseteq S$. If H did not imply it, then $X \subseteq T$ for some $T \in \max(H[S], A)$, and $T \to A$ would follow from $X \to A$ by augmentation, a contradiction. With $H \subseteq \Sigma^+$, the two closures contain the same FDs over S. Whether S is in 3NF depends on the FDs over S alone, so S is in 3NF with respect to Σ as it is with respect to H by Lem. 2.3, and $\Sigma[S] = H[S]$, since both consist of the FDs over S that are minimal in the common closure. □

The careful mode of Sec. 5 settles conflicting answers as follows. A net no to an FD that H implies is a *conflict*. It is settled by asking again the question whose majority is smallest, among the no and a minimal set of yes answers that imply the FD. A changed net answer rebuilds H, the refutations and the maximal sets, and the loop continues. The certificate is issued when no pair needs a question, no question is unanswered, and every premise has a majority of at least two answers or the questions of a conflict have all been asked three times.

Proof of Prop. 5.5. If every premise is right, then $H^+ \subseteq \Sigma^+$, since H is generated by Σ_0 and the yes premises, and every maximal set of H is false, since the no premises refute those that hold on r and r, which satisfies Σ, refutes the others; hence $H^+ = \Sigma^+$ by

Table 8: The Relations with Data: Attributes, Rows, Candidates, and True Candidates under the Full/Sure Truth; (A) Marks Class A

development										confirmatory				
relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true
food	17	312,847	78	31/7	tpcds-customer	25	100,000	68,156	44/44	auctionmark (A)	22	34,743	4,243	36/15
hospital	16	1,000	145	54/22	ldbc-comment	25	1,000,000	1,784	26/26	seats (A)	24	77,399	4,640	292/19
pricepaid	15	926,485	34	13/4	ldbc-study	24	7,949	1,249	39/39	tatp (A)	18	376,152	909	28/28
ssb-customer	8	3,000	24	12/10	gtfs	24	1,000,000	1,170	23/23	simplewiki (A)	17	999,656	139	36/30
ssb-date	17	2,557	209	205/113	faa	25	317,999	695	14/14	chembl	25	999,405	385	204/43
ssb-lineorder	16	600,572	3,814	13/5	abs	17	368,286	78	59/59	musicbrainz	25	1,000,000	85	71/61
ssb-part	9	20,000	86	3/3	sec	25	999,991	428	16/15	pluto	25	858,284	1,572	202/46
ssb-supplier	7	200	21	11/9	scorecard	25	160,030	5,169	64/33	fhrs	22	579,855	1,488	25/25
tpcds-core	12	2,685,687	48	23/13	epa	23	381,023	1,895	25/13	geonames	25	1,000,000	2,900	12/12
tpch-lineitem	23	600,572	36,883	10/10	pums	25	393,725	2,056	820/8	eia-860	25	27,919	1,097	14/14
tpch-orders	20	150,000	484	44/44	chicago	22	259,629	314	21/13	stats19	24	128,272	1,670	42/42
tpch-partsupp	19	80,000	2,405	25/25	usda	21	644,125	161	69/25	ghcn-daily	19	999,369	351	13/13

Thm. 4.2, and the design is good for Σ. The certificate is therefore wrong only if some premise is wrong, or if a conflict persists. In the latter case some net answer among the no and the yes answers that imply it is wrong, since the true FDs imply no false FD, and its question has received three answers. Think of the answers as drawn in advance, three for every question, each wrong independently with probability ε, of which Cert3NF reads the first one, two or three, since no question receives more than three answers. In either case a net answer that is a majority of at least two answers is wrong, which requires at least two of the three drawn answers of its question to be wrong, with probability $3\varepsilon^2(1-\varepsilon)+\varepsilon^3 = 3\varepsilon^2-2\varepsilon^3$. Number the different questions in the order in which they are first asked. Whether a j-th question is asked, and which one, depends only on the answers read before its first answer, which are independent of its own drawn answers. Hence, the probability that a j-th question is asked and two of its drawn answers are wrong is $\Pr[Q_r \geq j](3\varepsilon^2-2\varepsilon^3)$, where Q_r is the number of different questions of the run, and the sum over j is $Q(3\varepsilon^2-2\varepsilon^3)$. Within a budget of 100 answers, $Q \leq 100$, and $100(3\varepsilon^2-2\varepsilon^3) < 300\varepsilon^2$. With an error rate ε_q per question, drawn independently of the other questions, the same argument applies with the probability $3\mathrm{E}[\varepsilon_q^2]-2\mathrm{E}[\varepsilon_q^3]$ that two of the three drawn answers are wrong. □

Proof of Prop. 5.6. The data refute an FD only if it fails on r up to θ, so the FD is not in Σ^+. For $X \subseteq M$, every group of records that agree on M lies within a group that agrees on X, so the records that remain after the removal for $X \to A$ also satisfy $M \to A$. Hence, $M \to A$ holds up to θ whenever $X \to A$ does, and (a) follows as in the proof of Thm. 4.4 with a minimal $X \subseteq M$ for which $X \to A$ holds up to θ. A set of answers that certifies a design refutes every maximal set of H that holds on r up to θ, and the other maximal sets are refuted by the data and hence false, so the design is good for Σ by Thm. 4.2, which does not refer to the data. The proofs of (b) and (c) use the data only in this way, to refute the maximal sets whose FDs fail on r, and argue about closures otherwise, so they carry over unchanged. Part (d) does not, since the FDs that hold on r up to θ need not be closed under implication. □

B EXPERIMENTAL DETAILS

This appendix gives the protocol of Sec. 6. The artifact holds the implementation, the mined relations with their ground truth and every raw result.

Environment. Synthesis, Cert3NF and the estimators were implemented in Python. The LLM is Qwen3.5-9B in FP8, served by vLLM 0.31.0 on one NVIDIA RTX 4090 in each of two servers, and every other computation ran on the 16 CPU cores and 60 GB of memory of these servers. Times are wall-clock per run and include the advisor, but not the mining of the candidates and the answers of the model, which are computed once beforehand. The second model of (Q3) is gpt-oss-20b in its released MXFP4 weights with medium reasoning effort, served the same way and asked the same prompts, and the error rates of its advisor are estimated from its answers on the development relations, as for Qwen. The model answers one prompt per candidate, which names the attributes, states Σ_0 and shows a few rows, with yes or no and a confidence. Every candidate of every relation is answered once, with a fixed seed per prompt, and Cert3NF reads the answer of a candidate only when it queries the model. The normalizer runs an evidence, a diagnosis and a generator agent of the same model with up to 32,768 output tokens per call, and a deterministic verifier that checks the design against the inferred FDs and returns it for at most two repairs.

Random schemata and TPC-H. For the random schemata we drew Σ over n attributes with 2 to $n+1$ FDs, whose left-hand sides have one to three attributes, 300 times for each $n \in \{4,5,6\}$ and 150 times for $n = 7$. Into each we injected, one at a time, up to 12 random false FDs $X \to A \notin \Sigma^+$ with $|X| \leq 2$ and every missing FD of the canonical cover. This gave 10,737 false and 2,958 missing FDs for $n \leq 6$, and 1,800 and 655 for $n = 7$. On TPC-H, the true FDs follow from the specification: besides the primary keys, the names of customers, suppliers, nations and regions are keys, the brand determines the manufacturer, the country code of a phone number determines the nation, the part together with the quantity or the extended price determines the other, and the ship date determines the line status. HyFD discovers 3,945 minimal FDs on the eight tables, of which 76 follow from the specification, and we injected each of the 3,869 accidental ones alone.

Development relations. The first block of development relations in Tab. 8 consists of three denormalized joins that cover

the tables of TPC-H, the five tables of SSB, a core of TPC-DS, and the real data sets food (Chicago food inspections), hospital and pricepaid (UK land registry). Their ground truth is given by the specification of the benchmark or by the rules that the publisher documents. The relations of the second block were chosen and prepared by the rules below: a customer cluster of TPC-DS, the comments and the study relation of LDBC SNB, the GTFS feed of Auckland Transport, the FAA aircraft registry, the ABS statistical geography, SEC financial statement data sets, the College Scorecard, EPA air quality, the ACS PUMS of California, Chicago crimes and the USDA nutrient database. For every result on the development relations, the learned components of Cert3NF, the prior and the error rates of the model, were trained on the first twelve relations, leaving the relation under test out when it is one of them. For the confirmatory relations they are trained once on all twenty-four, and the kind of prior and of answer model were chosen by their held-out log-loss before any confirmatory answer was seen. The model has no answers for tpcds-customer, whose 68,156 candidates were not all answered, so the methods that use the model were run on the other 23 relations.

Confirmatory relations. The confirmatory relations were chosen from their documentation alone, before any data were downloaded, by rules fixed in advance: public, with a specification or data dictionary that states keys, code tables and hierarchies, outside the development relations and the collections used in related work, at least four benchmarks with a published schema (class A) and the rest real data (class B), at most two relations per publisher. Each relation was specified before its data were opened: which tables, which documented foreign keys to join along, which columns to keep, at most 25 columns, at least a thousand rows, and a sample of one million rows with a fixed seed where there were more. Class A consists of the items, flights and subscribers of the AuctionMark, SEATS and TATP benchmarks, generated by BenchBase at scale factor one, and the category links of the Simple English Wikipedia (MediaWiki). Class B consists of the activities of ChEMBL, the recordings of MusicBrainz, the tax lots of NYC PLUTO, the establishments of the UK Food Hygiene Rating Scheme (fhrs), the places of GeoNames, the generators of EIA-860, the casualties of STATS19 and the daily observations of GHCN-Daily. The ground truth is the projection of the documented FDs onto the chosen columns, each with a quotation from the documentation, written by a reader of the documentation who did not see the data. An FD that the documentation states explicitly is sure, and one that has to be inferred is not. The data were cleaned as the development relations were, and a greedy repair removed records until the sure FDs hold. It changed only chembl, fhrs, ghcn-daily and simplewiki and removed at most 631 records, except 5.7% of the records of fhrs, where one documented rule fails on the published data. On the unrepaired data of these four relations, the documented FDs fail on at most 0.09% of the records, except the rule of fhrs on 5.6% and, under the full truth, an inferred FD of chembl on 4.5% and one of simplewiki on 0.6%. Sec. 6.3 also reports Cert3NF on the unrepaired data, with and without the tolerances $\theta \in \{0.001, 0.01, 0.1\}$, which were fixed before any run with $\theta > 0$. After mining, a sample of at most 200 candidates per relation, 2,098 in all, was audited blindly against the documentation by readers who saw neither the ground truth nor any answer of the model. No documented FD was refuted, two candidates of GeoNames that the documentation implies were added, and three inferred FDs were promoted to sure. Before any method ran, we listed the documented FDs that the data violate, seven inferred FDs on seats, simplewiki, chembl, musicbrainz and pluto, and checked that every documented FD that holds on the data is minimal there.

Simulated designer. The documented FDs and the declared key play the designer: a question whether $Z \to A$ holds is answered yes if and only if they imply it, and a question that the data refute is not asked. For (Q4), the simulated designer answers wrongly with probability ε, independently on every answer. Whether an answer is wrong is decided by a hash of the relation, the seed, the question, ε and the number of times the question was asked, so that every method meets the same errors. A designer who *repeats* an error gives the first wrong answer again when asked again.

Pre-registered rules. Before the first confirmatory relation was evaluated, we wrote the following rules into the design record of the project, judged per ground truth over the relations, with the two classes reported separately. (C1) Every certificate of Cert3NF is correct: the checker verifies its premises against the ground truth, and its design is good, except on the relations listed before the run where the data violate a documented FD. (C2) Paired by relation, Cert3NF certifies more relations than Cert3NF without the LLM, or as many with at least three fewer answers in the median where both certify, under both truths. (C3) Cert3NF certifies at least as many relations as the certification phase without an advisor, and needs at most b plus one answer per confirmed maximal set more than it on at least 90% of the relations. (C4) Over the certified runs, the median ratio of answers to the price of Thm. 4.4 is at most 2, and no certificate has fewer no answers than m_r. (C5) At the end of the budget, Cert3NF has at least as many good designs as the posterior certificate, counted per relation by the median of its three seeds. The rules were judged once. A later fix of an unstable sort, which makes Cert3NF deterministic, changed one cell within the budget and no verdict, and the tables report these recomputed runs. C1, C4 and C5 hold on all relations, on class A and on class B under both truths. C4 was registered with a price computed from the true candidates, which counts a held set once for every attribute for which it is maximal and takes the smallest of the covers that four elimination orders leave. On the confirmatory relations, this price is never below that of Thm. 4.4, which the tables report and under which C4 holds as well. We find a smallest F exactly, by integer programming over the left-reduced FDs of Σ. Where the data violate a documented FD, F is a smallest cover of the FDs of Σ^+ that hold on the data, and the held sets are the maximal sets of Σ that contain the left-hand side of a false candidate. C3 holds on all relations and on class A, and misses class B under both truths by fhrs alone. C2 holds on class B under both truths and on all relations under the sure truth, but not on class A or on all relations under the full truth. Since it requires both truths over all relations, we present the model as an optional part of the advisor. The normalizer and the second model of (Q3) ran after these verdicts, on the same relations. The posterior certificate and the EC^2 certificate draw 80 samples from the posterior, and the posterior certificate accepts a design at $\delta = 0.1$. The last rule concerns the EC^2 certificate, which weighs the samples of the posterior equally, assigns each sample to the first design that covers it in a greedy cover of the samples

Table 9: Answers until the Certificate on the Development Relations, as in Tab. 3; ✗ Marks a Wrong Certificate, and tpcds-customer Has No Answers of the Model (n/a) and No Run without a Budget

		full truth						sure truth					
relation	cand.	true	price	Cert3NF	w/o LLM	w/o advisor	posterior	true	price	Cert3NF	w/o LLM	w/o advisor	posterior
food	78	31	33	79†	76†	(152†)	3† (75)	7	11	26	25	49	3✗✗✗ (55)
hospital	145	54	36	(112)	(144)	(121)	3✗✗✗ (96)	22	34	91	81	99	3✗✗✗ (80)
pricepaid	34	13	15	34†	35†	55†	3† (32)	4	7	17	11	26	3 (33)
ssb-customer	24	12	11	28	28	25	3 (18)	10	10	23	13	21	3 (19)
ssb-date	209	205	23	27	76	(154)	3 (40)	113	30	71	73	(118)	3 (70)
ssb-lineorder	3,814	13	44	70	(108)	86	–	5	33	38	48	48	2 (85)
ssb-part	86	3	11	14	13	20	3 (35)	3	11	14	13	20	3 (40)
ssb-supplier	21	11	10	21	24	22	3 (15)	9	9	23	12	20	3 (15)
tpcds-core	48	23	17	26	36	78	3✗✗✗ (25)	13	25	72	66	80	3✗✗✗ (23)
tpch-lineitem	36,883	10	49	72	54	96	2✗✗ (57)	10	49	75	77	96	1✗ (80)
tpch-orders	484	44	32	56	52	(151)	1 (99)	44	32	62	52	(151)	1 (90)
tpch-partsupp	2,405	25	36	53	54	(108)	–	25	36	53	54	(108)	–
tpcds-customer	68,156	44	39	n/a	91	–	n/a	44	39	n/a	89	–	n/a
ldbc-comment	1,784	26	40	63	65	(208)	–	26	40	63	65	(208)	–
ldbc-study	1,249	39	46	79	88	(212)	–	39	46	76	88	(212)	–
gtfs	1,170	23	50	68	(103)	(167)	–	23	50	82	(103)	(167)	–
faa	695	14	38	50	52	98	3 (80)	14	38	48	52	98	3 (95)
abs	78	59	27	53	82	(156)	3 (73)	59	27	39	47	(156)	3 (67)
sec	428	16	35	44	69	(103)	3 (92)	15	37	69	82	(109)	2✗✗ (100)
scorecard	5,169	64	62	(103)	(297)	(161)	–	33	66	(200)	(220)	(152)	–
epa	1,895	25	124	(449)	(409)	(202)	–	13	125	(438)	(431)	(194)	–
pums	2,056	820	278	(616†)	–	–	–	8	26	29	34	61	1 (100)
chicago	314	21	32	55	60	(115)	2 (100)	13	26	43	39	74	3 (95)
usda	161	69	41	(144)	(165)	(154)	2 (90.5)	25	31	57	58	(125)	3 (70)
within 100				18/23	17/24	8/24	40/69			21/23	21/24	12/24	43/69

by good designs, and asks per round about the five candidates of largest expected weight of cut edges, with ties broken by the largest marginal. (C6) At the end of the budget, Cert3NF has at least as many good designs as the EC^2 certificate, counted as in C5, and certifies at least as many relations within the budget as the EC^2 certificate does in two of its three seeds. We fixed C6 after the evaluation above but before the EC^2 certificate ran, and it holds on all relations and on both classes under both truths, also against the EC^2 certificate without the LLM.

Nine designers. The experts of the study are database researchers and administrators from outside our group, and its practitioners are graduate students and developers who have taken a database course and designed schemata. Every person worked on two relations in two sessions one to seven days apart, so that in every condition each relation comes first once and second once. A session starts with a fixed block of six questions about candidates, six about maximal sets and six about other subsets, half of them true (on abs only nine questions, all true), and then runs Cert3NF in the careful mode with $\theta = 0$. A question about a maximal set names the attributes outside it: must any two records that agree on all attributes except B_1, B_2 and A also agree on A? The registered claim that maximal sets are as answerable as candidates, with at most five points more errors at the upper end of the 95% interval, holds for the experts, who erred on neither, and is not established for the practitioners, who erred on 8% (17%) of the maximal sets against 3% (12%) of the candidates with (without) the documentation. The registered claim that errors are mostly independent, with the upper end of the 95% interval of $\hat{\phi}$ below 0.25 for the experts and for the practitioners with the documentation, does not hold for the latter, since $\hat{\phi}$, estimated from the first two answers to each repeated premise, is 0.7 for them against 0.1 for the experts. Every expert passed the required check of at least five right answers to the six candidate questions of a block. An answer of don't know counts towards the budget, is not an error, and leaves its candidate or maximal set undecided. We kept the totals per condition, so that, unlike the registration, we report no intervals clustered by participant and no results per session, and leave out the registered comparison with the errors of the LLMs and the effects of question size and session order.

Attribute exploration. We run exploration by NextClosure [15] on the confirmatory relations, with the attributes in their order in the relation, the agree sets of the data as the known objects, Σ_0 as background knowledge, and the simulated designer as the expert. The classical expert refutes an implication by the smallest counterexample, the closure of its premise under Σ, and we also let it return the largest, a maximal set of Σ that contains the premise. To compare with Cert3NF, an implication becomes one yes or no answer per attribute of its conclusion that the confirmed implications do not yet imply, and the largest counterexample one answer per attribute by which it is grown, unless the data or the confirmed implications decide it. With the smallest counterexamples, exploration poses from about 15,000 to over three million implications,

and three relations under the full truth and four under the sure truth do not finish within two hours. With the largest, it poses a median of 28 implications, fewer than the price, since one yes confirms a whole conclusion and one counterexample is a maximal set, and Sec. 6.3 reports the answers they amount to.

Synthetic relations. To see whether the answers follow the price beyond the 25 columns of the real relations, we generated denormalized joins of snowflake schemata whose true FDs hold by construction. The fact key of order and line is declared, the entities form a tree under the fact, each with a key, one to four descriptive attributes and, with probability 0.3, a name that is a second key, and the remaining attributes are measures that only the fact key determines. Cardinalities and domains are drawn log-uniformly, values uniformly, and the attributes are renamed in a random order. One sweep varies the number of attributes at a share of 0.6 of the other attributes on the entity side, and the other varies the share at 30 attributes, each with three seeds of 20,000 rows. Before generating any data, we fixed that Cert3NF without the LLM and without an advisor certify every design within 1,000 answers (S1), that the median ratio of answers to the price is at most 2 in every configuration (S2), and that Spearman's ρ of the answers is larger with the price than with the number of candidates (S3). All three hold (Tab. 5 in Sec. 6.3). Beyond 30 attributes, the limit lies in mining and in holding the candidates: two of the three relations with 40 attributes are mined within two hours by our miner, with 5.6 and 6.9 million candidates, which take more than the 60 GB of memory of our servers, and none with 50 attributes is.